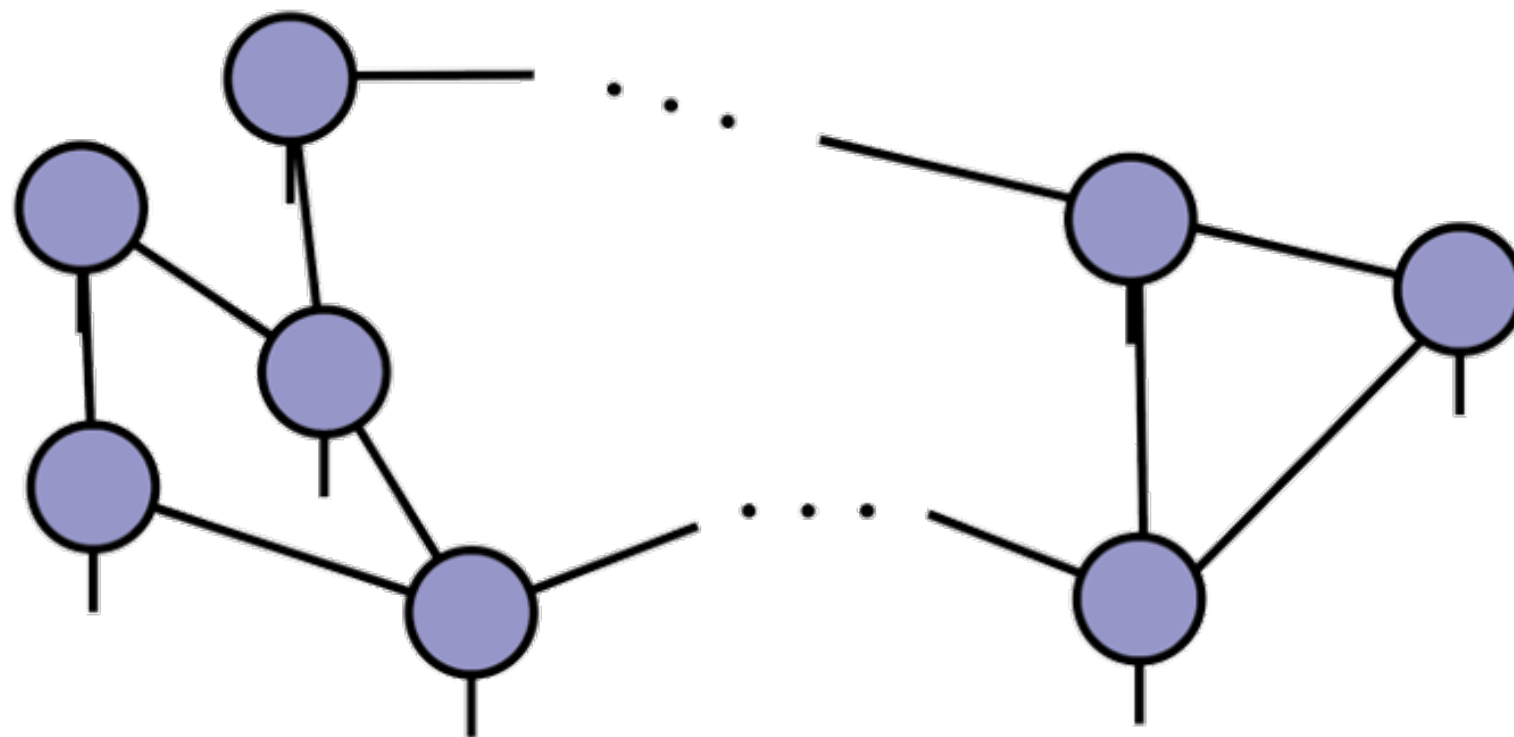
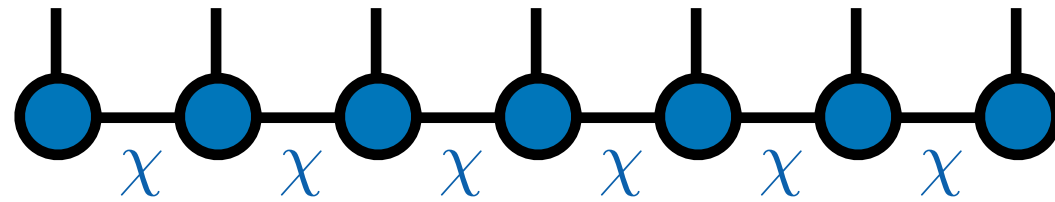


Tensor Networks Beyond One Dimension



Motivation

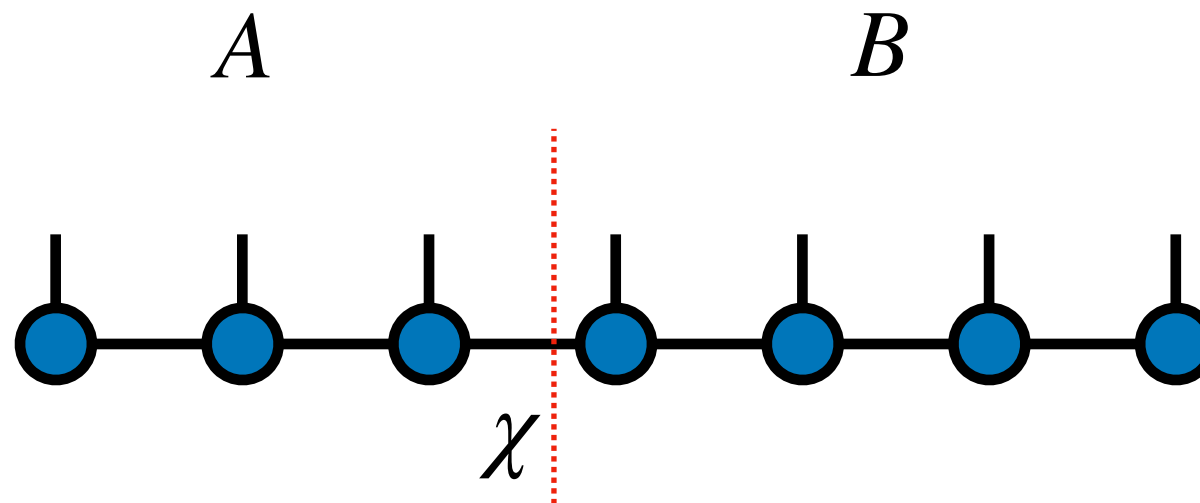
Earlier introduced tensor networks, but entirely focused on **matrix product states**



An MPS has a one-dimensional structure, making it powerful for **1D** but weak for **2D** and **3D** systems

What tensor networks can we use in higher dimensions?
Alternatives to MPS?

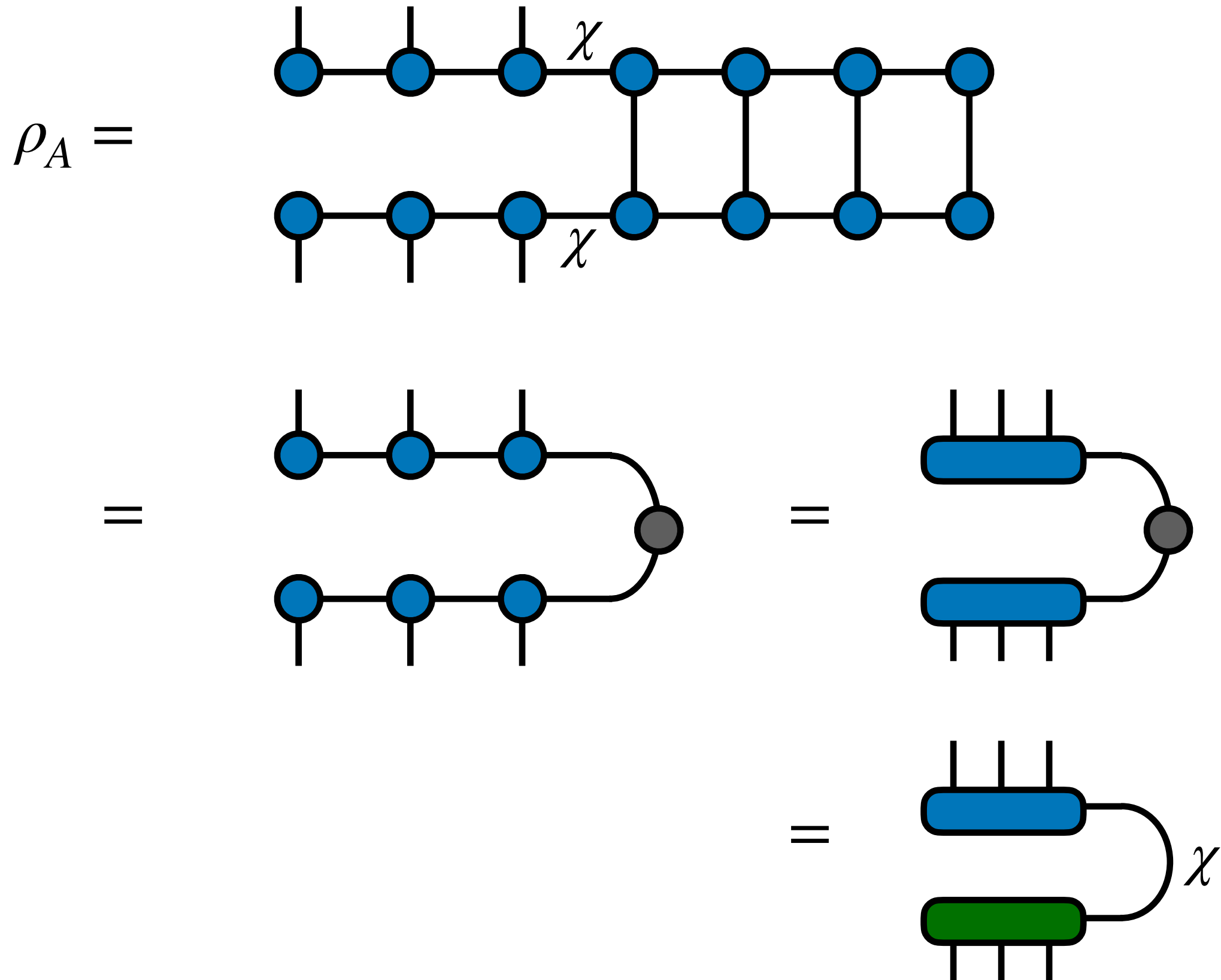
What is the entanglement capacity of an MPS tensor network?



Consider bipartition ("cut") across bond of dimension χ

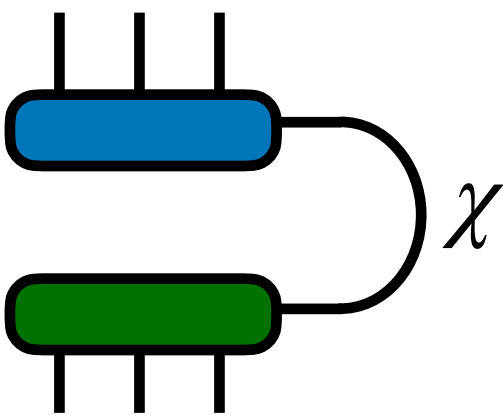
Compute reduced density matrix ρ_A

Entanglement capacity of an MPS



Entanglement capacity of an MPS

We have shown that ρ_A is a product of two matrices, contracted over an index of dimension χ

$$\rho_A = \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array}$$


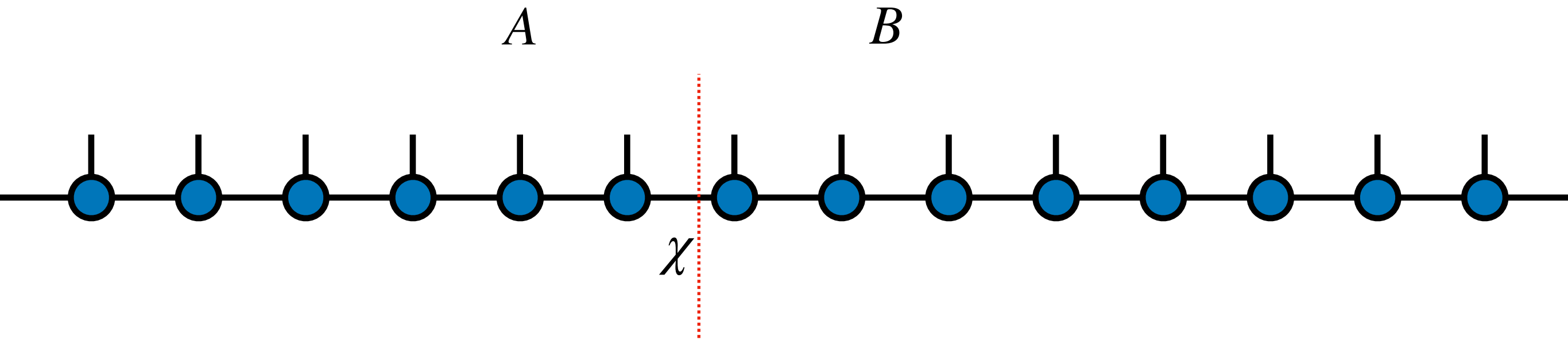
The diagram illustrates the contraction of two matrices, represented by a blue rectangle and a green rectangle, over an index of dimension χ . The blue rectangle has three vertical lines extending upwards from its top edge, and the green rectangle has three vertical lines extending downwards from its bottom edge. A curved line connects the right side of the blue rectangle to the right side of the green rectangle, with the label χ next to it, indicating the dimension of the contracted index.

Thus ρ_A has maximum rank χ , i.e. at most χ non-zero singular values

Recall $S_{vN} = -\text{Tr}(\rho \log(\rho))$

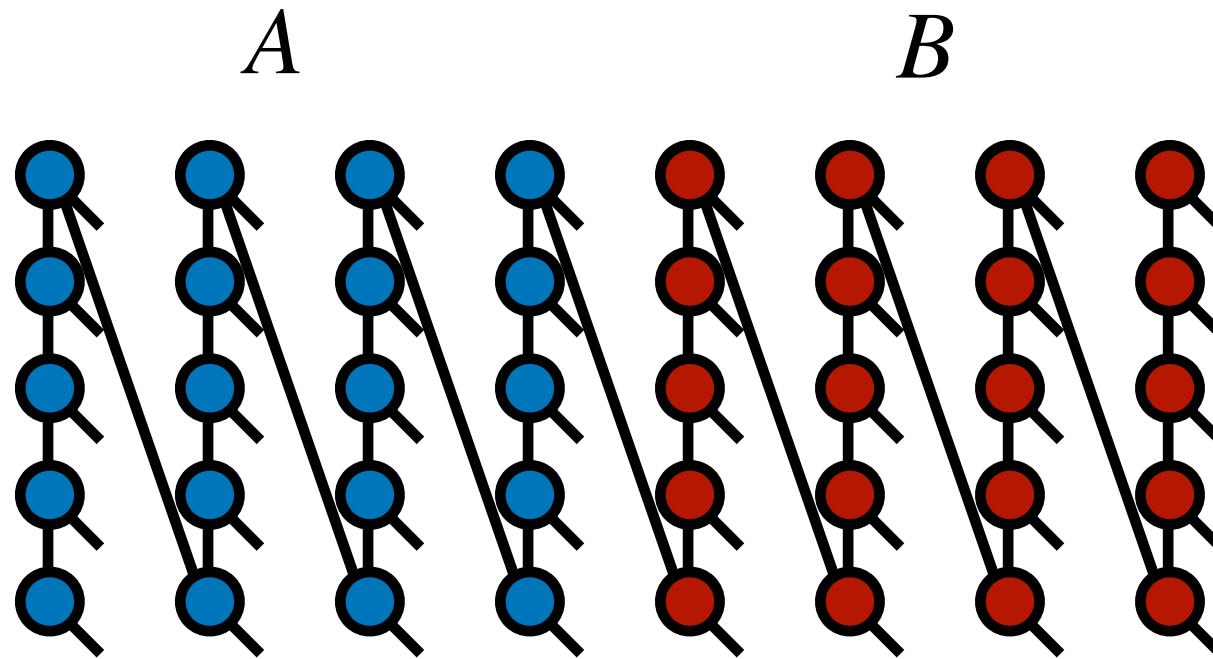
Maximum entropy if all eigenvalues $p_n = 1/\chi$, implies an upper bound $S_{vN} \leq \log(\chi)$

We therefore see MPS with $\chi \sim \mathcal{O}(1)$ obey a 1D area law



Entropy bounded $S_{vN} \leq \log(\chi) \sim N^0$ (i.e. const with system size)

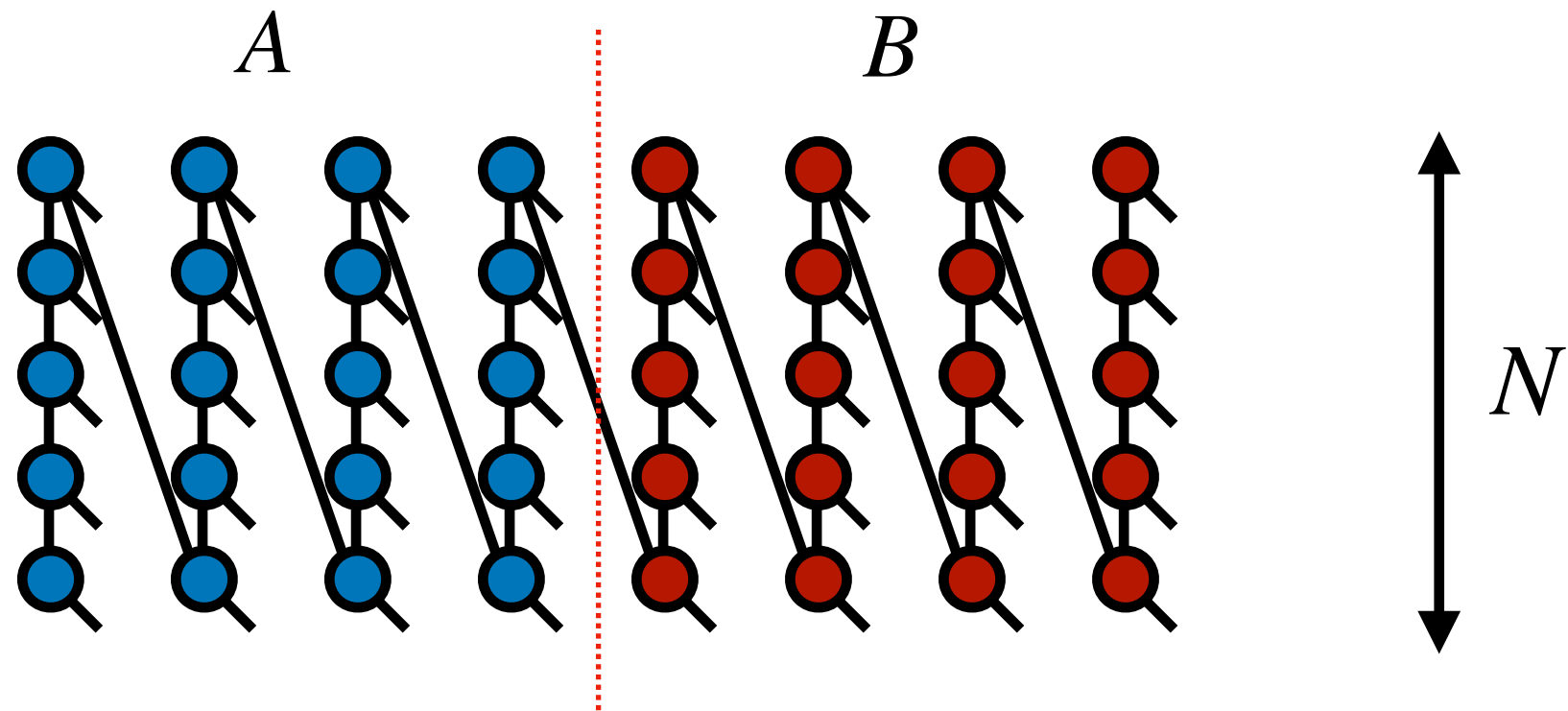
If we use MPS for a two-dimensional system (2D)



we can treat the 2D system using a "zig-zag" or "snaking" pattern to visit all the sites

Basis of 2D-DMRG approach that has been successful on *thin* 2D systems

But importantly, scaling of entanglement is still that of an MPS

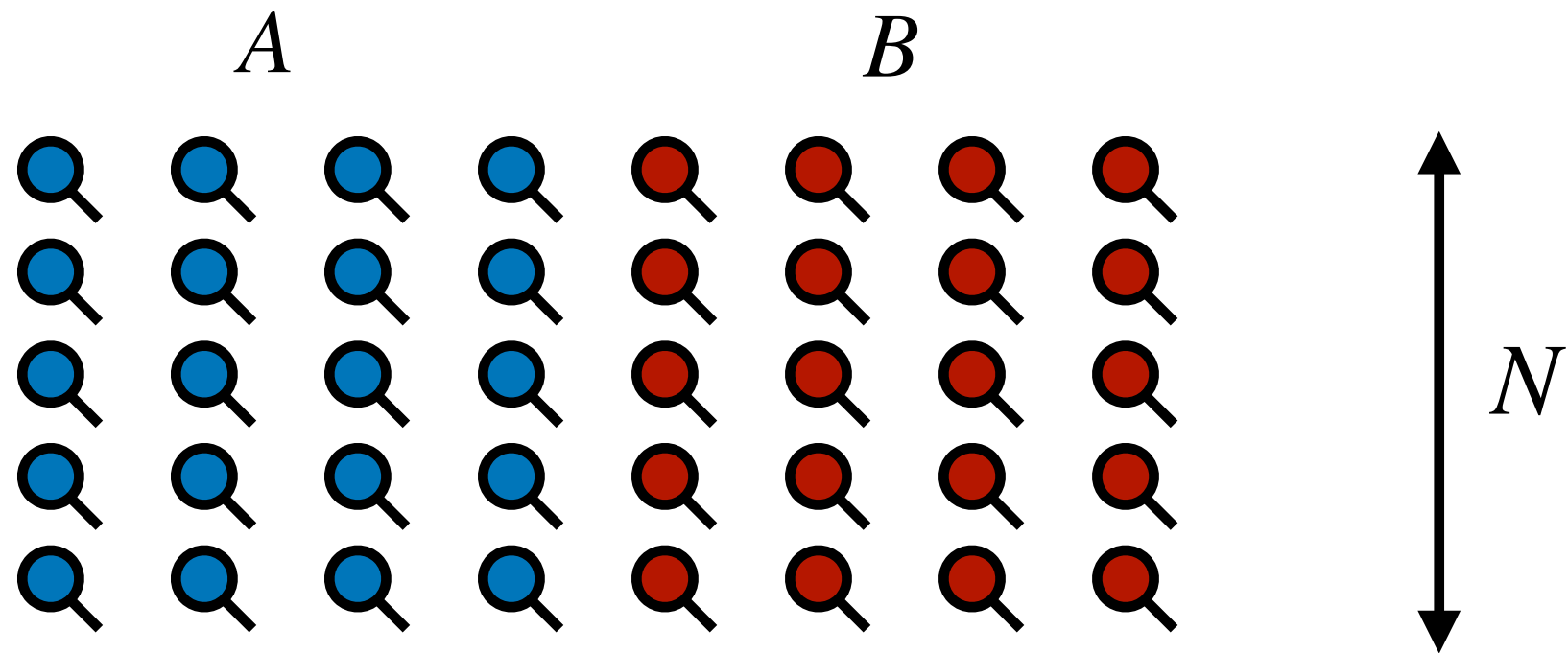


$$S_{vN} \leq \log(\chi)$$

If we assume 2D wavefunction is "nice" and obeys 2D area law, where $S_{vN} \sim aN$

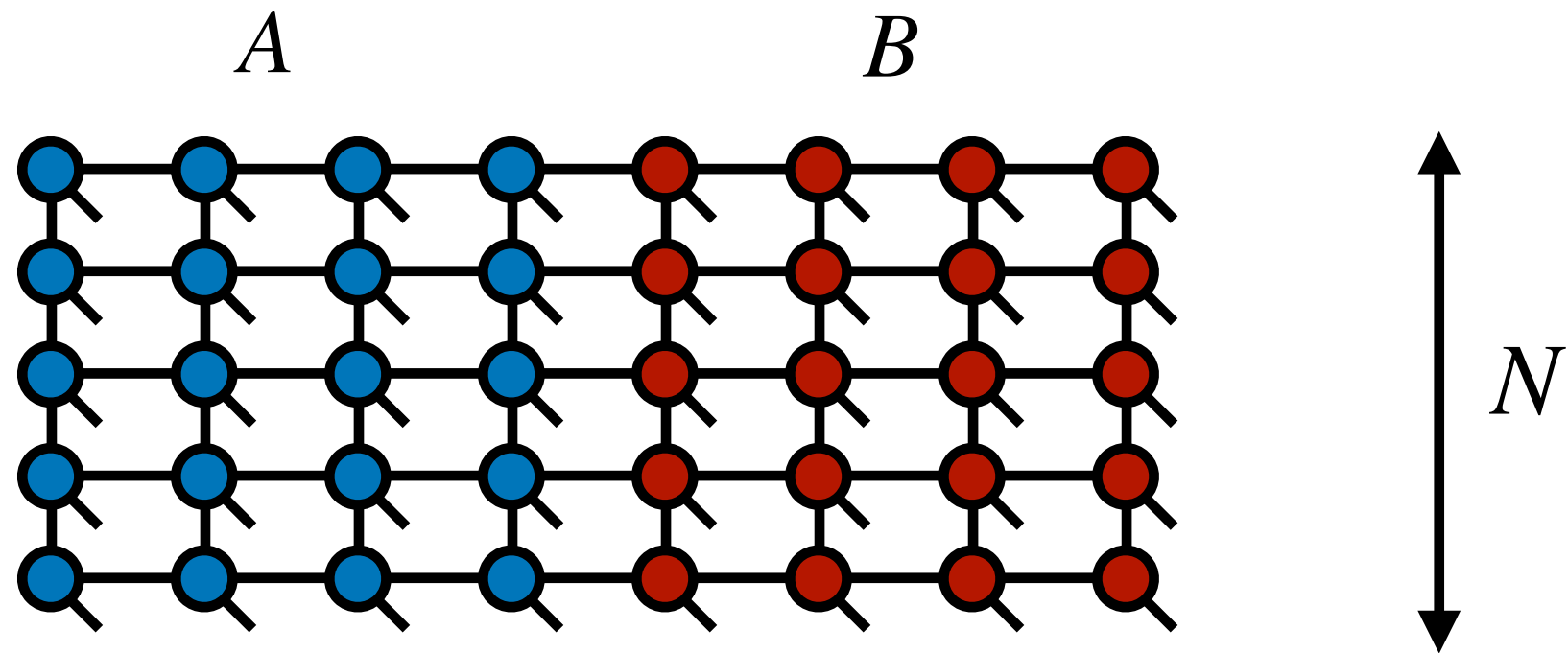
Then requires $\chi \sim e^{aN}$ to capture the state

Intuition: each bond of size χ contributes $\sim \log(\chi)$ entanglement



Want total $S_{vN} \sim N$, so put N bonds across the cut (i.e. connecting regions A and B)

Intuition: each bond of size χ contributes $\sim \ln(\chi)$ entanglement



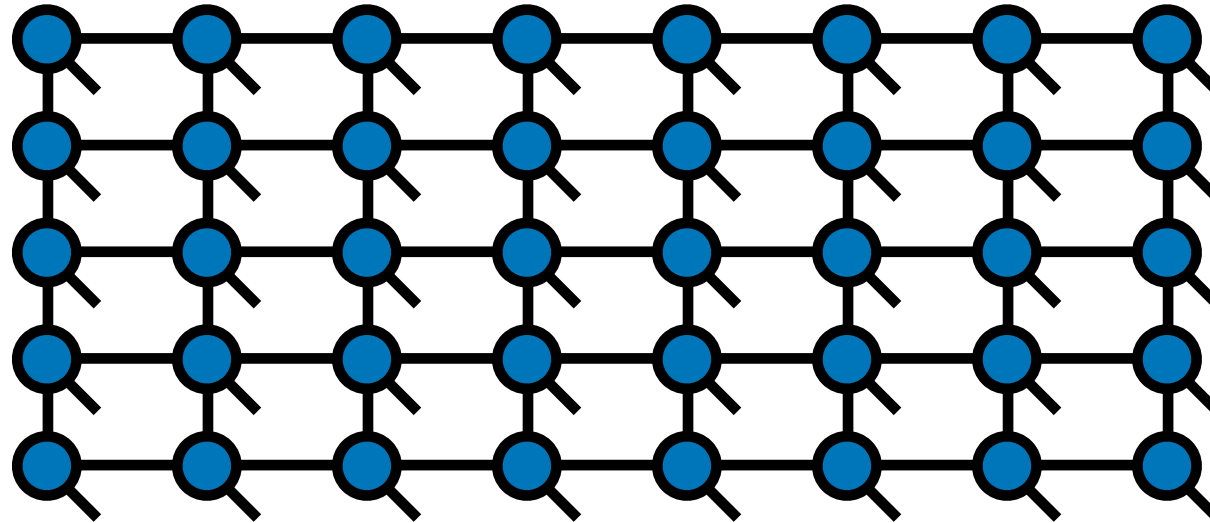
Want total $S_{vN} \sim N$, so put N bonds across the cut
(i.e. connecting regions A and B)

Motivates 2D tensor network state (TNS)
also called projected entangled pair state (PEPS)

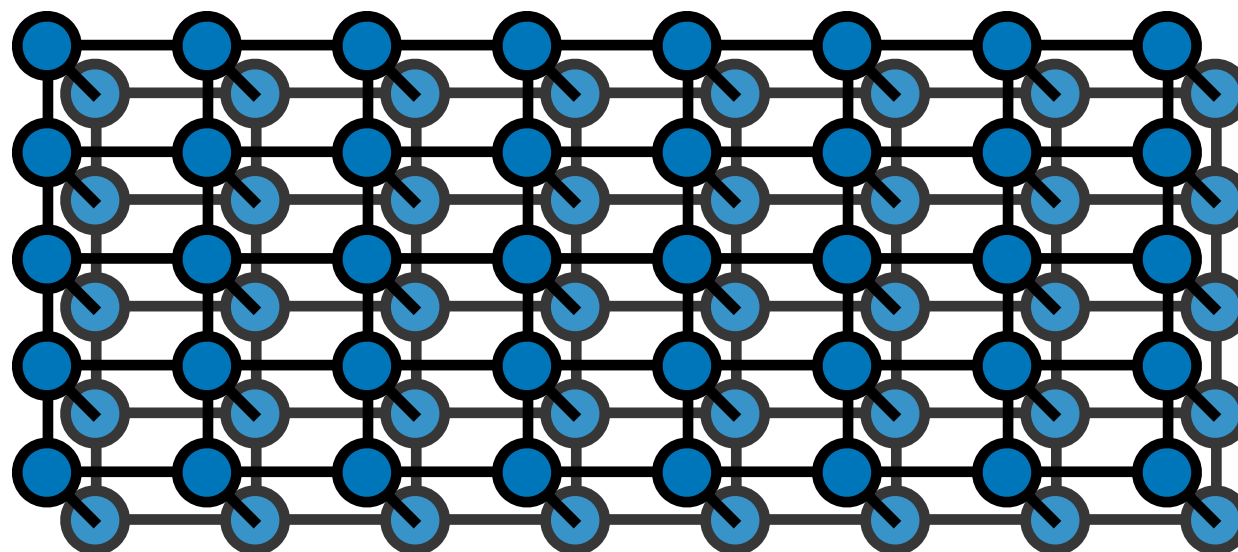
The Belief Propagation Algorithm for General Tensor Networks

Motivation

To work with 2D tensor networks

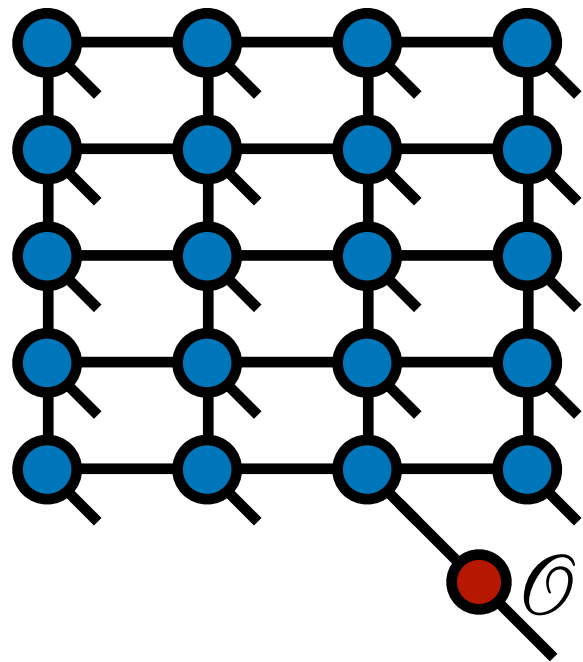


Must be able to contract them

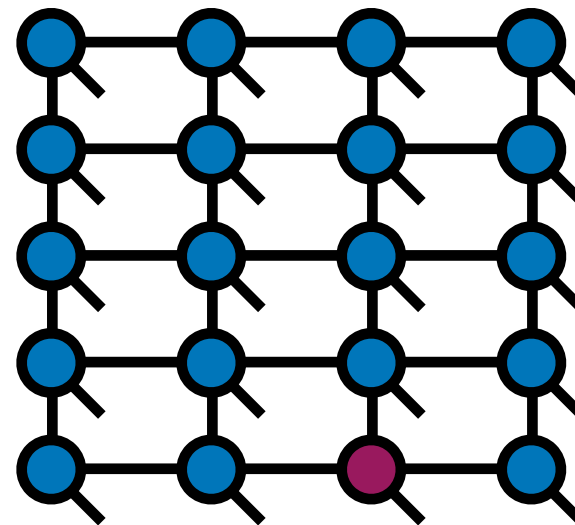


Motivation

For example, computing operator expected values

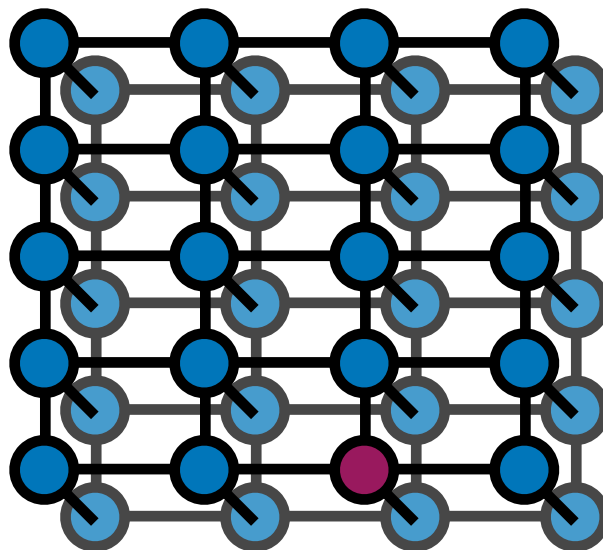


=



$\langle \psi | \mathcal{O}$

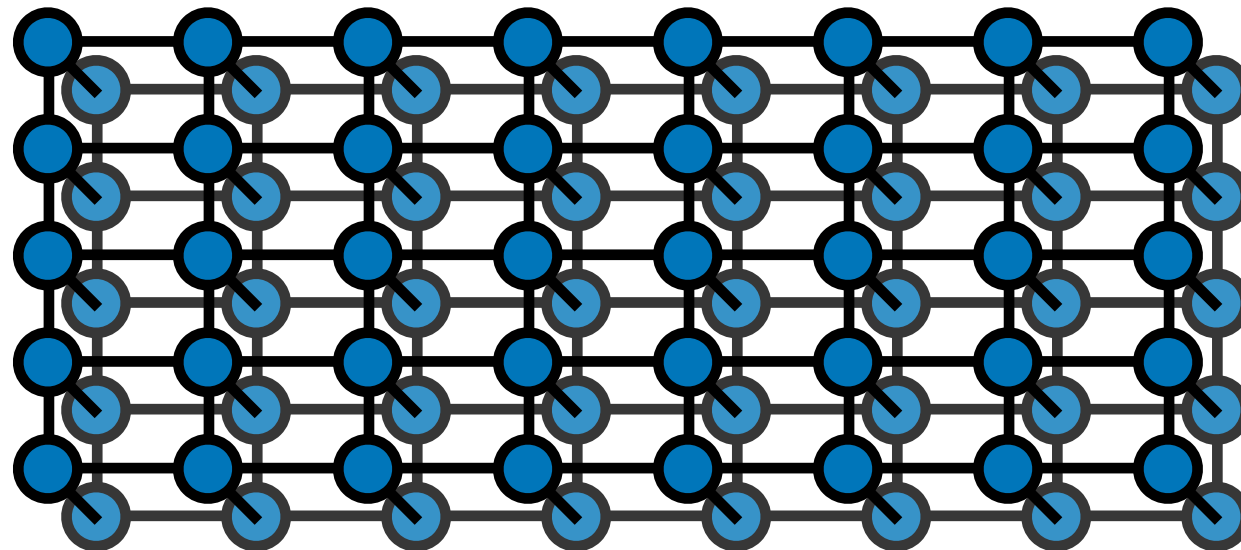
$\langle \psi | \mathcal{O} | \psi \rangle =$



Motivation

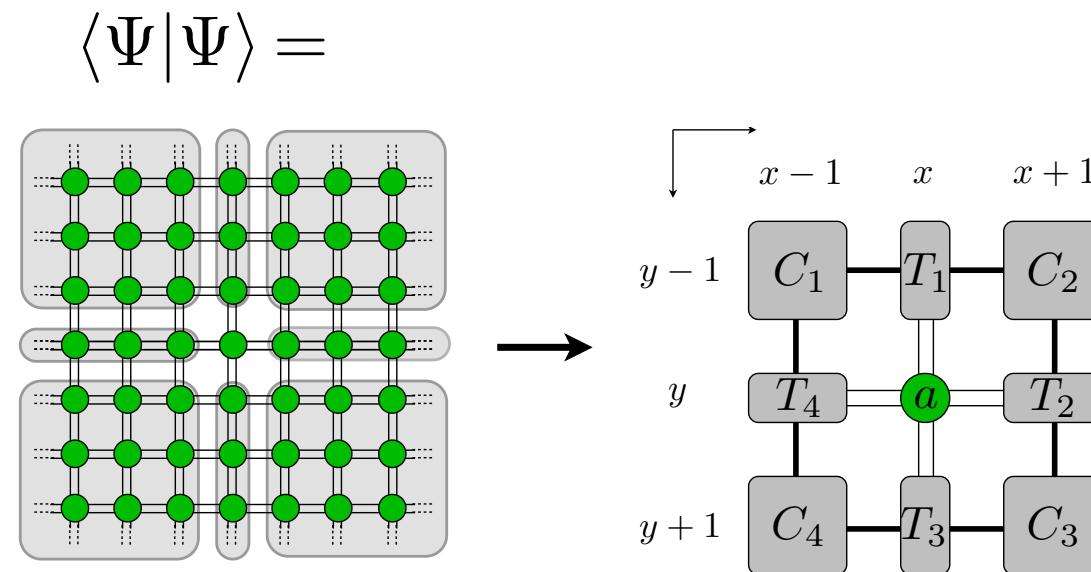
However, contracting 2D (3D, etc.) tensor networks exactly is #P-Complete in worst case.

A polynomial time (in system size) to contract any finite bond dimension PEPS would imply $P = NP$.



Motivation

Highly controlled approximations have been developed for *infinite PEPS* (TNS) optimization [1,2]



and related works using automatic differentiation [3,4]

[1] Corboz, Phys. Rev. B **94**, 035133 (2016)

[2] Vanderstraeten, Haegeman, Corboz, Verstraete, Phys. Rev. B **94**, 155123 (2016)

[3] Liao, Liu, Wang, Xiang, Phys. Rev. X **9**, 031041 (2019)

[4] Ponsioen, Assaad, Corboz, SciPost Phys. **12**, 006 (2022)

Motivation

However, existing 2D tensor network methods are **expensive** and overly-focussed on square lattices

Typical costs: *

- iPEPS optimization – scaling $O(D^{10})$ for square lattice
- VMC optimization – scaling $O(MD^6)$ with $M \sim 10,000$ the number of samples

Also, **3D** lattices seem out of reach

* Liu, Zhai, Peng, Gu, Chan, arxiv:2502.13454

Motivation

Is there *more efficient* method for contracting TNS?

We need to accept *larger approximations...*

We just need *to know* how large the approximation is

 "Belief propagation" + extensions gives all three



Belief propagation has been around for some time

Hans Bethe

statistical mechanics (1935)

Statistical Theory of Superlattices

By H. A. BETHE, H. H. Wills Physical Laboratory, University of Bristol
(Communicated by W. L. Bragg, F.R.S.—Received February 13, 1935)

Judea Pearl

probabilistic inference (1982)

REVEREND BAYES ON INFERENCE ENGINES: A DISTRIBUTED
HIERARCHICAL APPROACH(*)(**)

Judea Pearl
Cognitive Systems Laboratory
School of Engineering and Applied Science
University of California, Los Angeles

Mezard, Parisi, Virasoro

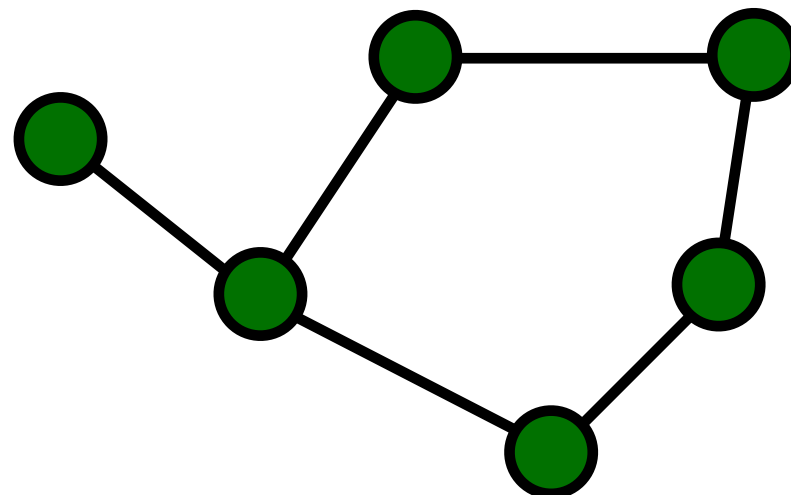
spin glass physics (1985)

J. Physique Lett. **46**, 217-222 (1985)
DOI: 10.1051/jphyslet:01985004606021700

Random free energies in spin glasses

M. Mézard, G. Parisi et M.A. Virasoro

Originally approximates *marginals* of
locally tree-like graphs (e.g. stat mech models)



Recently adapted to quantum wavefunctions via tensor networks

PHYSICAL REVIEW RESEARCH **3**, 023073 (2021)

Tensor networks contraction and the belief propagation algorithm

R. Alkabetz  and I. Arad

Department of Physics, Technion, 3200003 Haifa, Israel

[Submitted on 9 Jun 2022]

Efficient tensor network simulation of quantum many-body physics on sparse graphs

Subhayan Sahu, Brian Swingle

SciPost Physics

Gauging tensor networks with belief propagation

Joseph Tindall, Matt Fishman

SciPost Phys. 15, 222 (2023) · published 1 December 2023

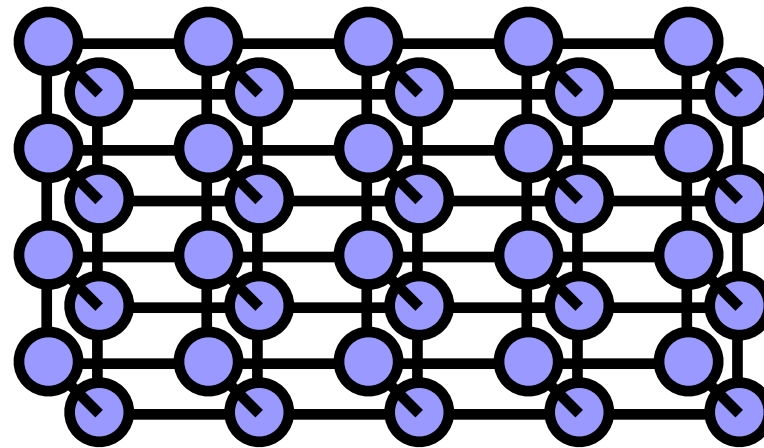
Alkabetz, Arad, Phys. Rev. Research 3, 023073 (2021)

Sahu, Swingle, arxiv:2206.04701

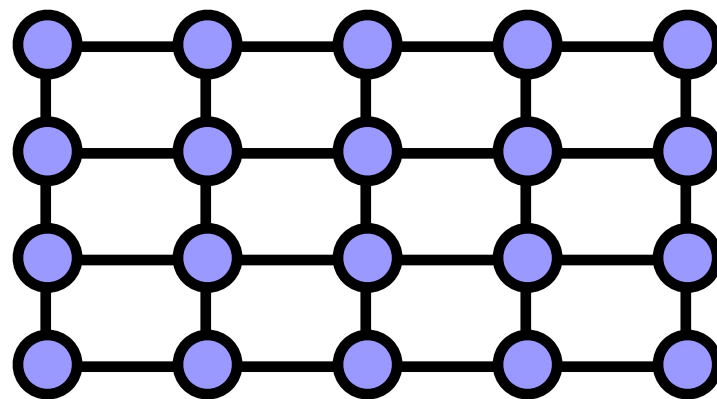
Guo, Poletti, Arad, Phys. Rev. B, 108, 125111 (2023)

Tindall, Fishman, SciPost Phys. 15, 222 (2023)

To apply belief propagation to quantum systems,
start from " $\langle$ bra | ket $\rangle$ " network ("norm network")

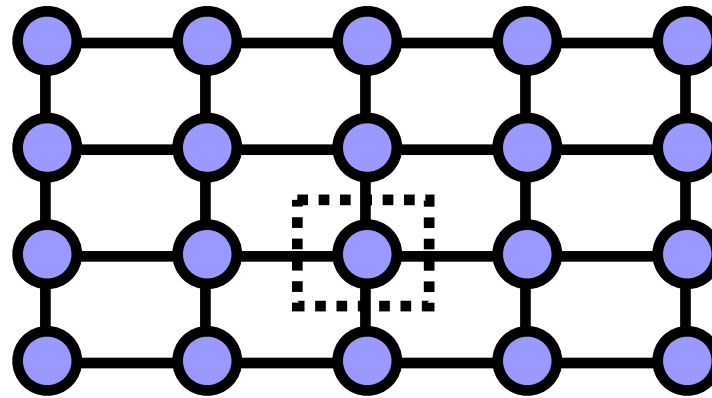


Top-down view



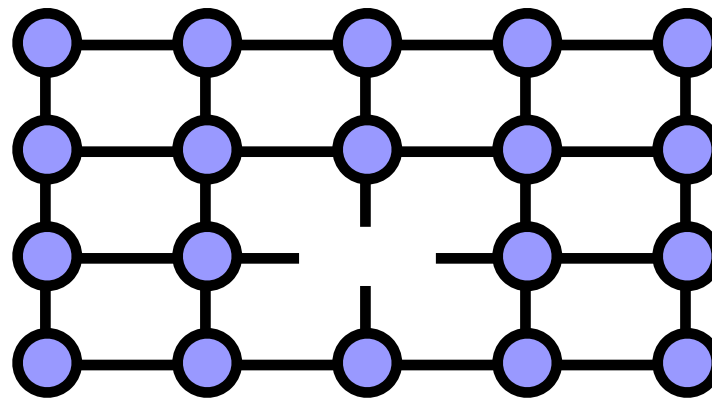
Ideally compute exact "environment"

Defined as network with one tensor removed

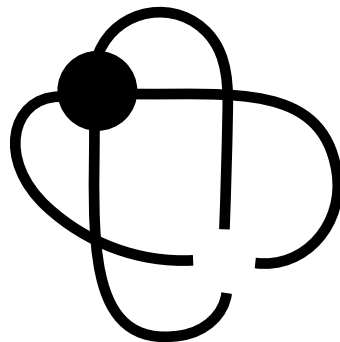


Ideally compute exact "environment"

Defined as network with one tensor removed

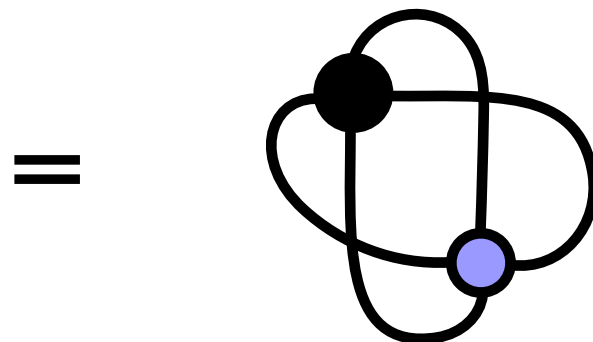
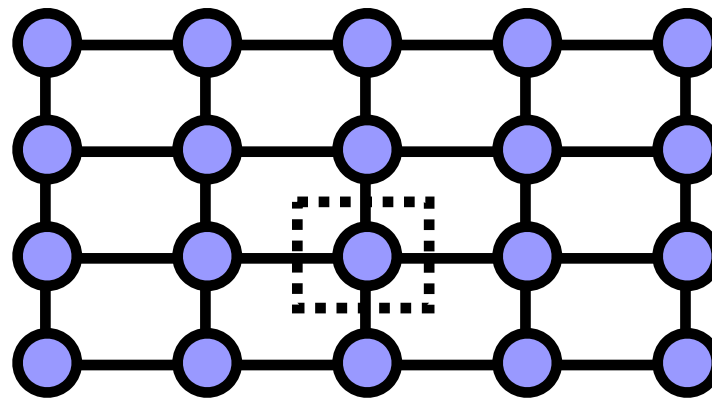


=



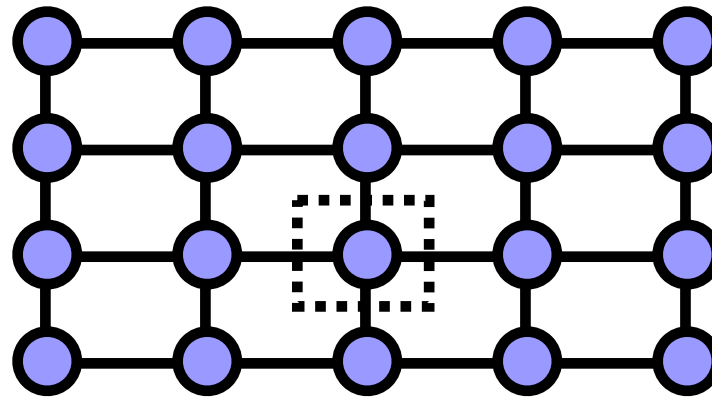
Ideally compute exact "environment"

Defined as network with one tensor removed

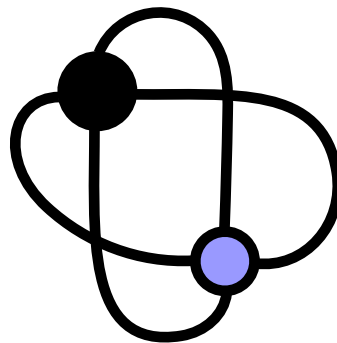


Small (4-index) environment tensor sufficient to contract whole network, but just as hard to compute

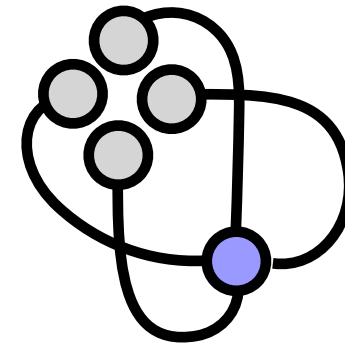
Make seemingly *drastic* approximation



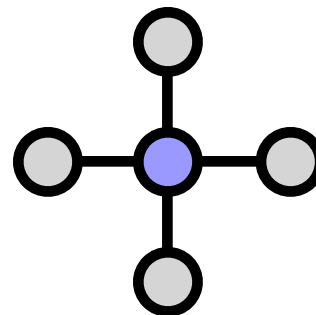
=



≈

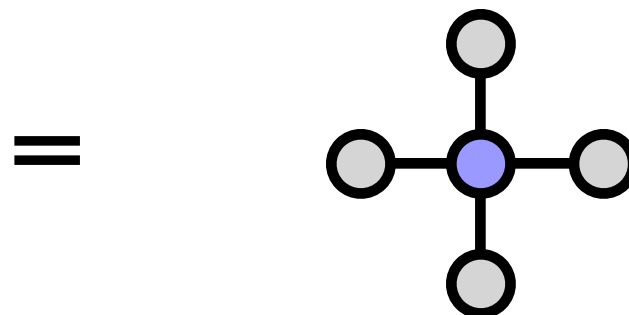
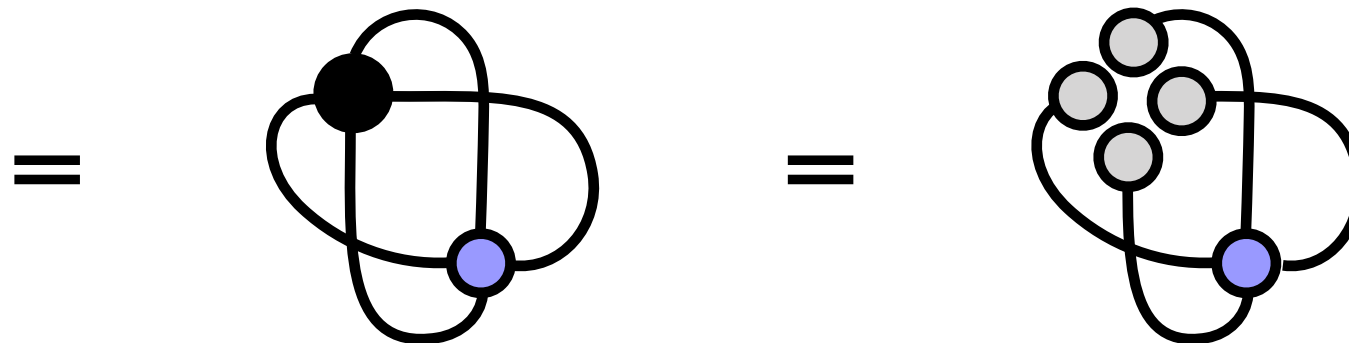
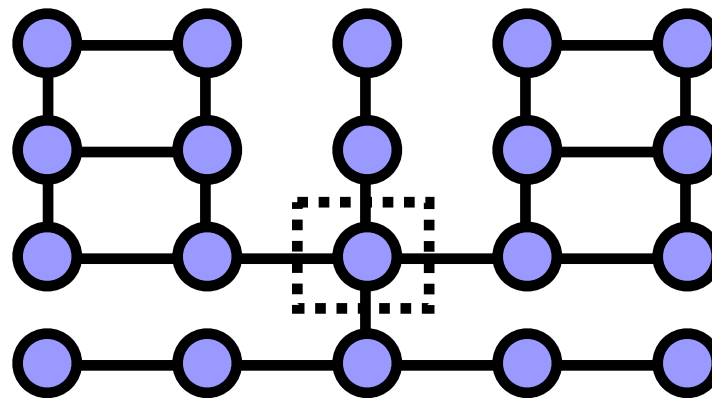


≈

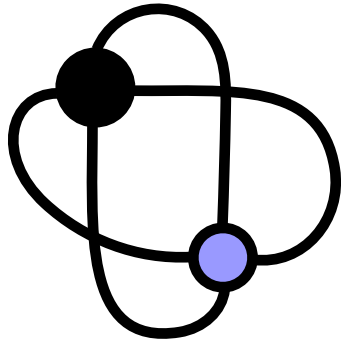


"messages"

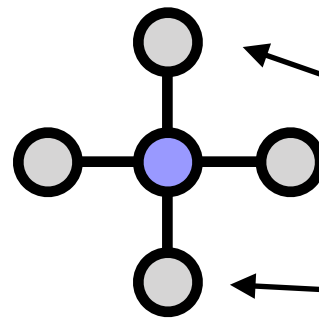
Would be exact if certain loops don't contribute



How to find "messages" in practice?



$\approx$

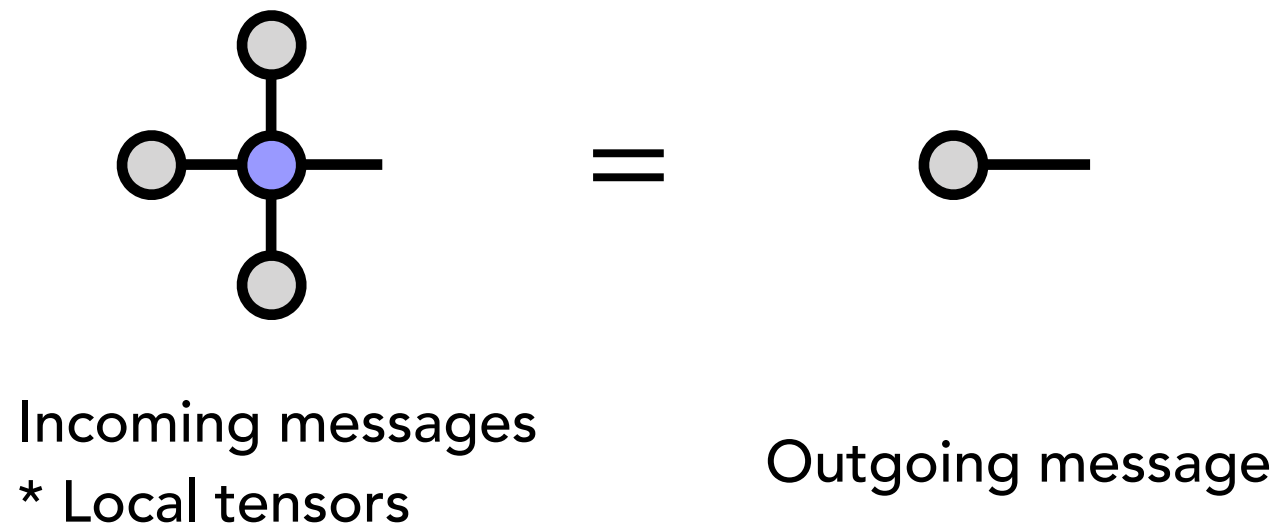


"messages"

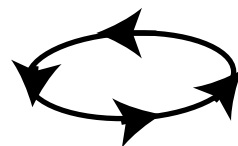
Message Passing

Use "message passing" to converge the messages until self-consistency over all edges of the network

Update rule on edge e



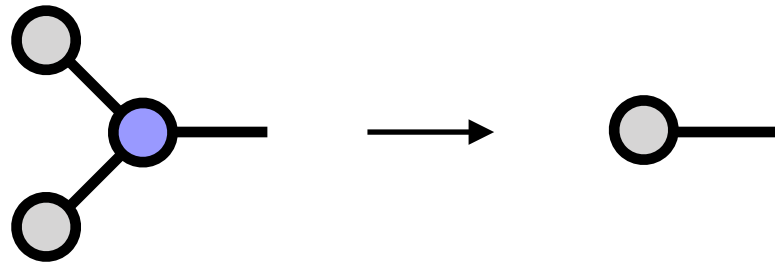
Iterate on all edges until convergence



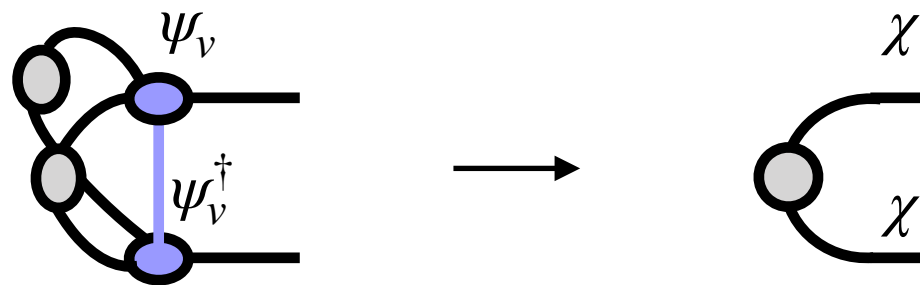
Belief Propagation

Side-on view helpful to understand quantum case. We typically run BP over the norm network $\langle \psi | \psi \rangle$

Top view:



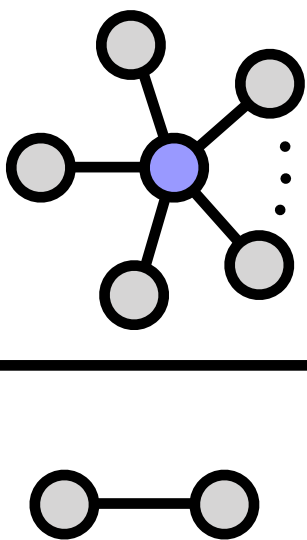
Side view:



Quantum messages are $\chi \times \chi$ matrices, where χ is the bond dimension. They are RDMS in the bond basis

Belief Propagation

After messages converged,
can compute BP estimate of norm ("partition function")

$$\langle \psi | \psi \rangle \approx \frac{\prod_{\text{vertices}} \text{diagram}}{\prod_{\text{edges}} \text{diagram}}$$


The diagram illustrates the Belief Propagation (BP) approximation for the norm of a tensor network, $\langle \psi | \psi \rangle$. The expression is shown as a ratio of two products:

- Numerator:** A product over all vertices, $\prod_{\text{vertices}}$. The diagram for a vertex is a central blue circle connected to five surrounding gray circles. A vertical ellipsis on the right indicates additional connections.
- Denominator:** A product over all edges, $\prod_{\text{edges}}$. The diagram for an edge is two gray circles connected by a horizontal line.

If the tensor network represents a partition function, the logarithm of this expression is exactly the Bethe free energy. If $|\psi\rangle$ is a tree tensor network, the approximation becomes exact.

Belief Propagation

Scaling of quantum belief propagation on the norm $\langle \psi | \psi \rangle$ is

$$O(N\chi^{z+1})$$

where z is the *coordination number* of the lattice, χ the bond dimension, N the number of sites.

So, for square lattice, scaling is $O(\chi^5)$

Compared to scaling and prefactors of other TNS algorithms, generally much better

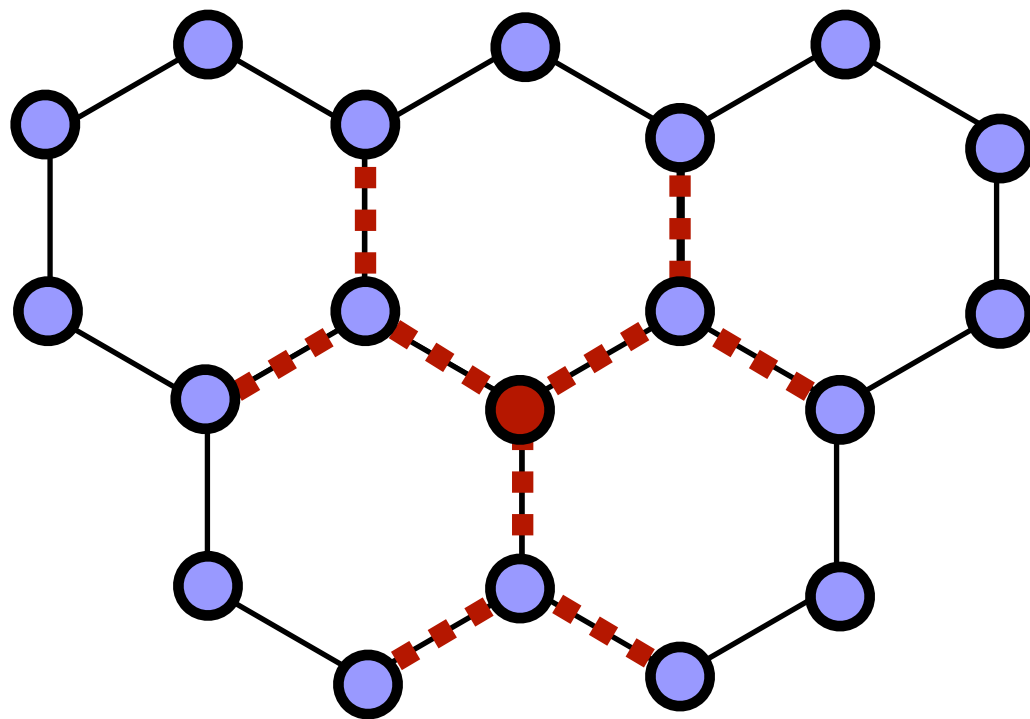
More neighbors (higher z) costs more, but
lower bond dimensions usually needed (mean field like)

BP Approximation

Intuitive understanding of BP approximation is viewing its convergence as

$$\epsilon_{BP} \sim e^{-a \frac{L_{\text{loop}}}{\xi}}$$

Correlation length



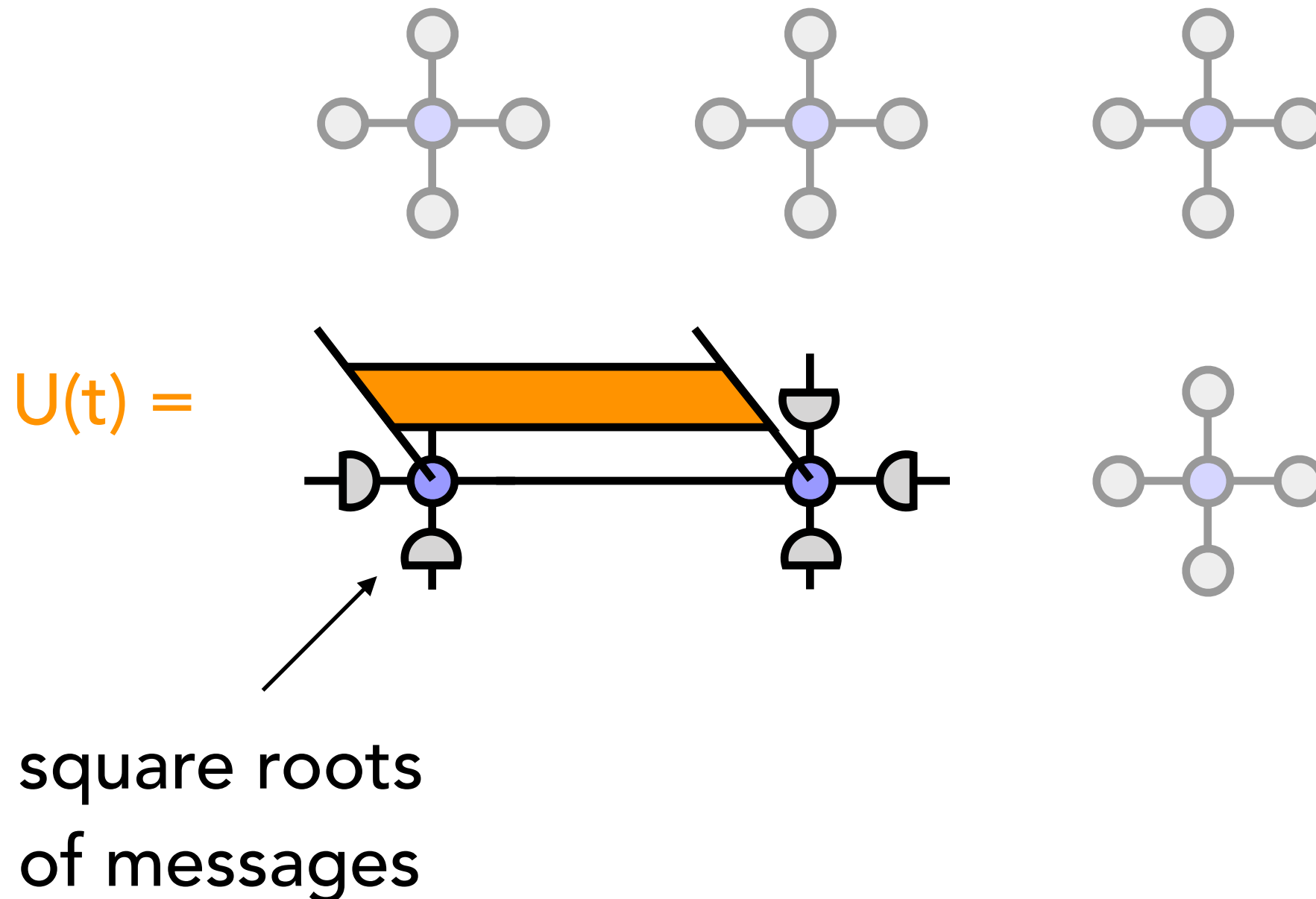
honeycomb lattice

$$L_{\text{loop}} = 6$$

If correlations do not transit around a loop,
loop might as well not be there

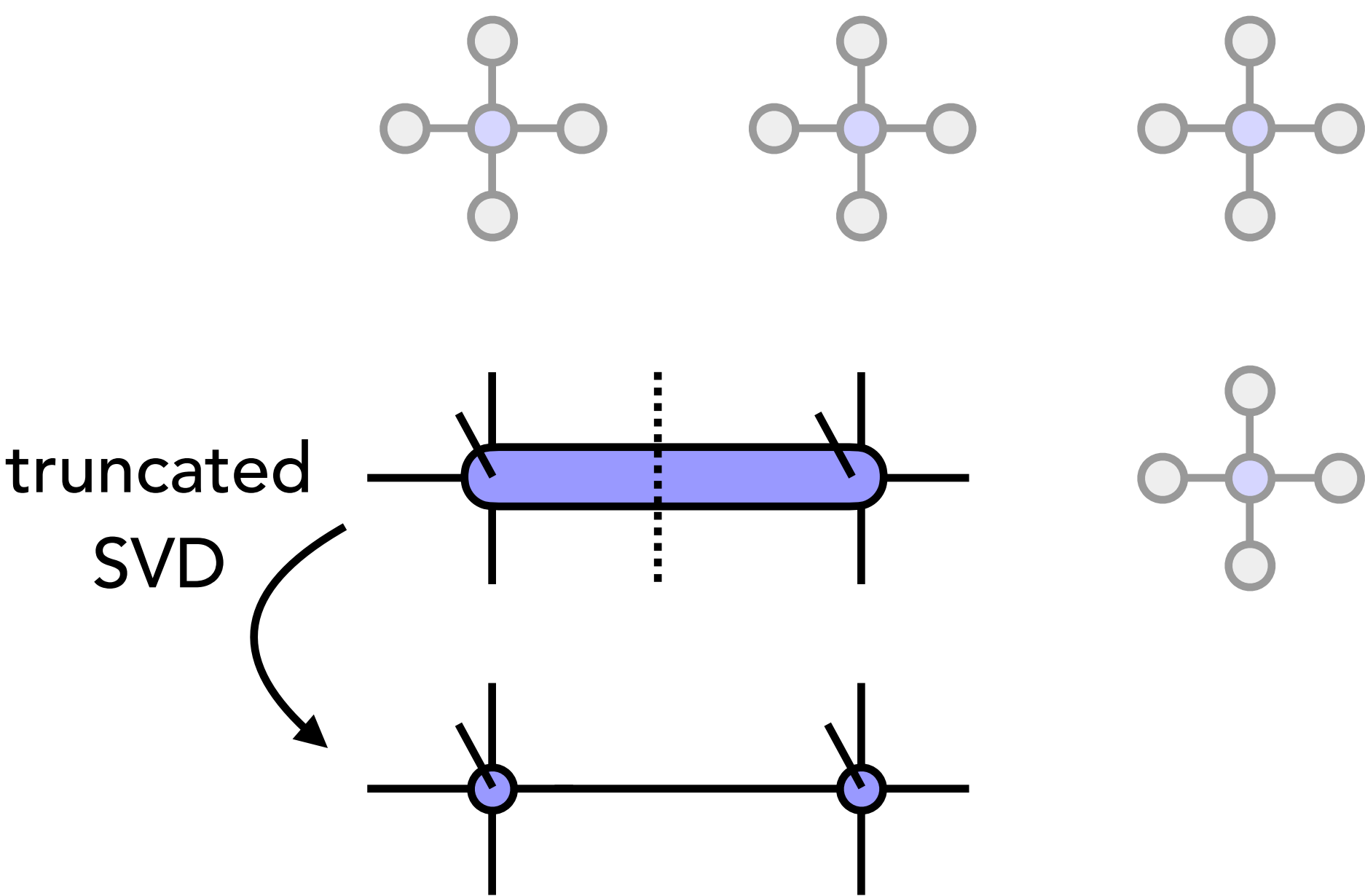
Belief Propagation

Use converged messages to apply gates



Belief Propagation

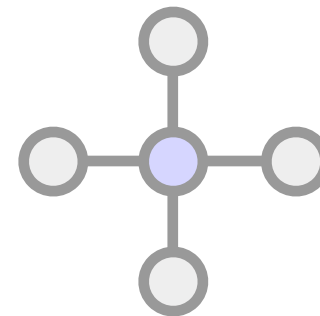
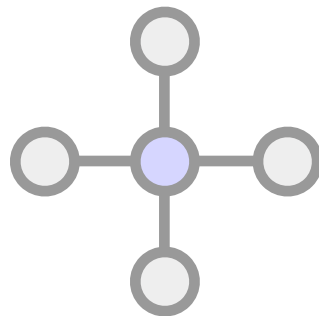
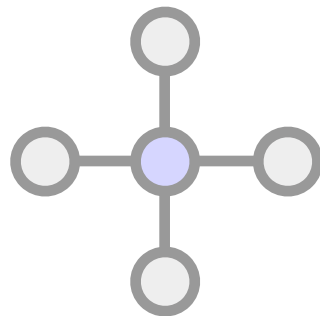
Messages play important role not so much when we apply the gate, but in order to make the SVD truncation accurate:



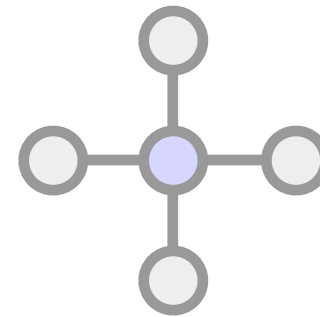
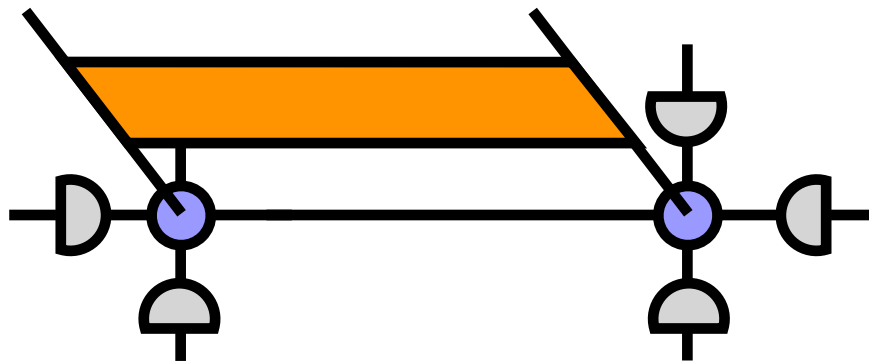
Belief Propagation

Applying gates lets us:

- 1) compute ground states & thermal states (imaginary time)
- 2) compute dynamics (real time)



$U(t) =$



Belief Propagation

For extensive discussion of theory of
quantum belief propagation
(relation to gauging, simple update, etc.)

SciPost

SciPost Phys. 15, 222 (2023)

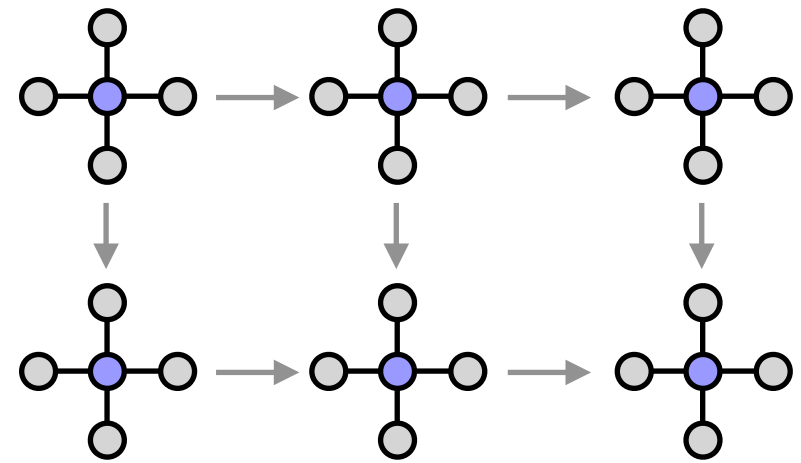
Gauging tensor networks with belief propagation

Joseph Tindall[★] and Matt Fishman

Center for Computational Quantum Physics, Flatiron Institute,
New York, New York 10010, USA

★ jtindall@flatironinstitute.org

Belief Propagation



Benefits of belief propagation:

- if lattice has no loops, fully **controlled**
- **cheap**: use large internal bond dimensions χ
- like mean-field theory, but more **accurate**
- can run **dynamics** on top
- **systematic corrections** possible

Applications of Belief Propagation for Quantum Simulation

1) Simulation of quantum utility experiment

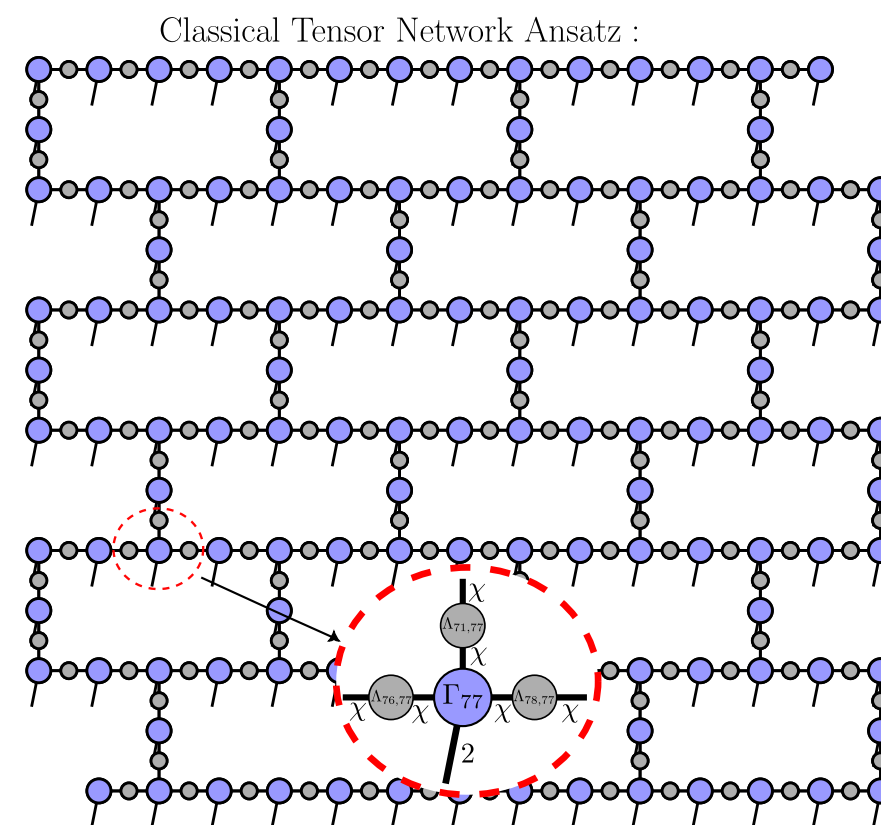
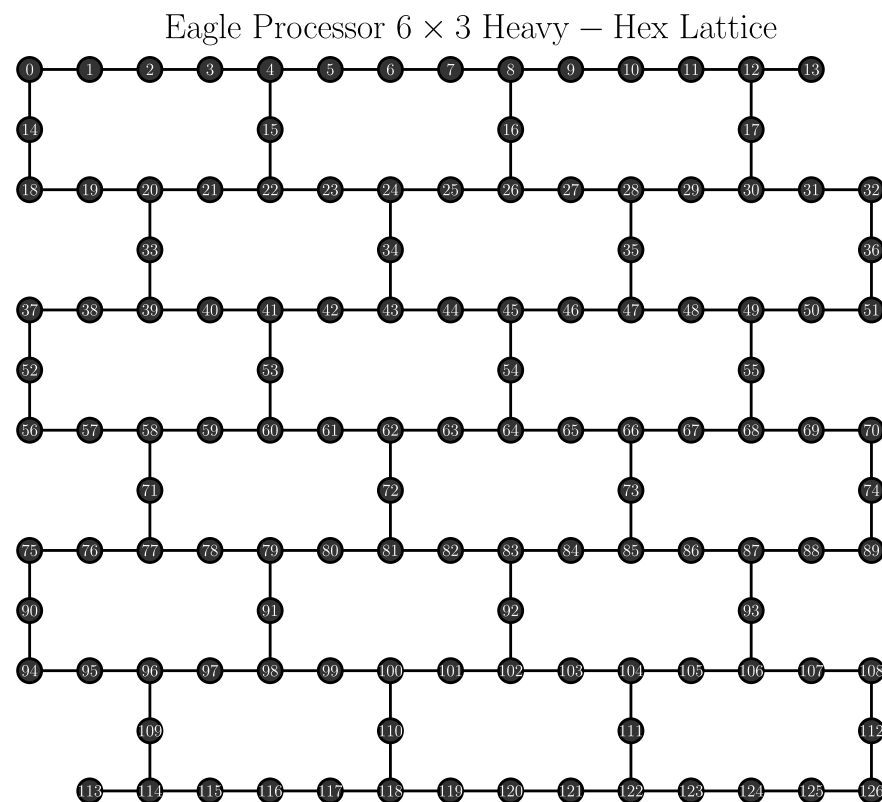
Featured in Physics

Open Access

Efficient Tensor Network Simulation of IBM's Eagle Kicked Ising Experiment

Joseph Tindall, Matthew Fishman, E. Miles Stoudenmire, and Dries Sels
PRX Quantum **5**, 010308 – Published 23 January 2024

Physics See Research News: [A Moving Target for Quantum Advantage](#)



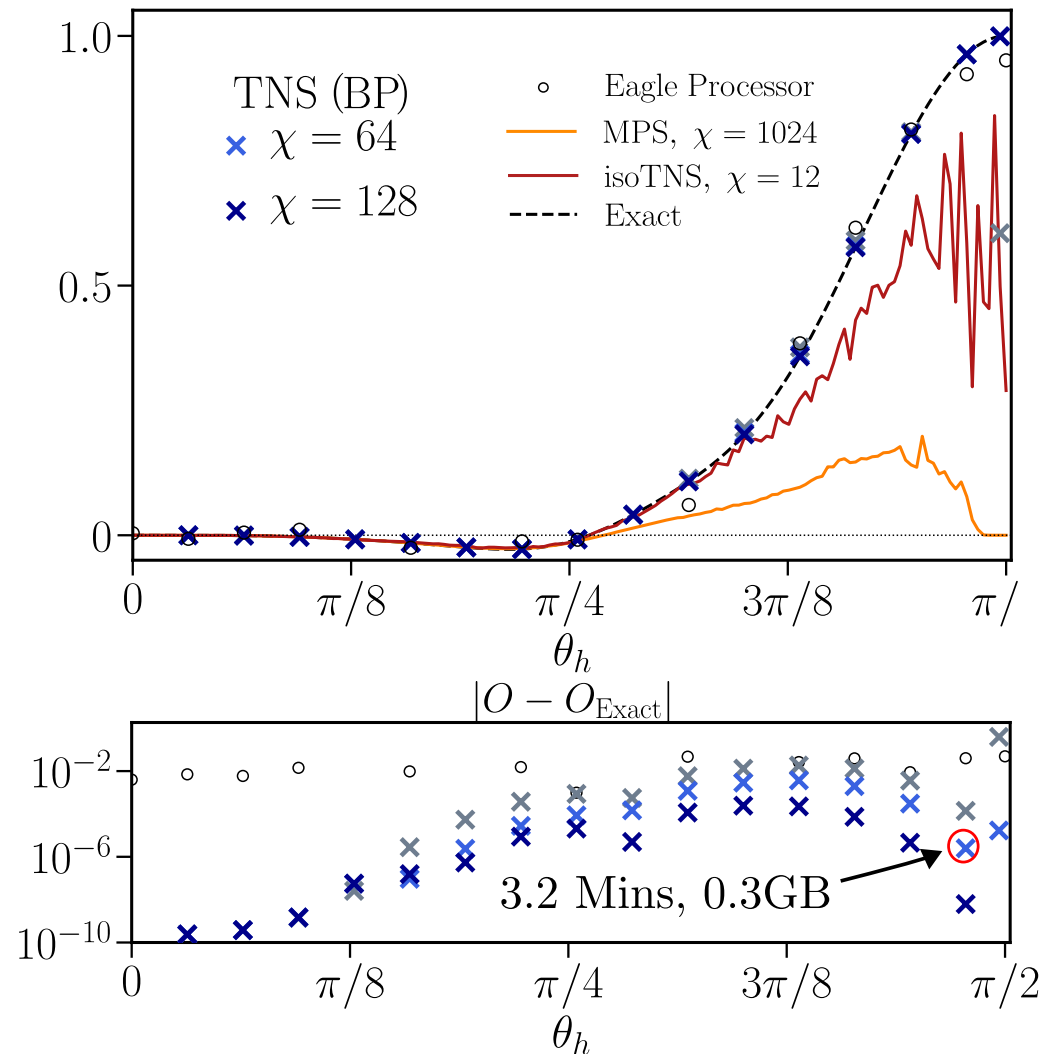
$$U(\theta_h) = \left(\prod_{\langle v, v' \rangle} \exp \left(i \frac{\pi}{4} Z_v Z_{v'} \right) \right) \left(\prod_v \exp \left(-i \frac{\theta_h}{2} X_v \right) \right)$$

← discretized dynamics

1) Simulation of quantum utility experiment

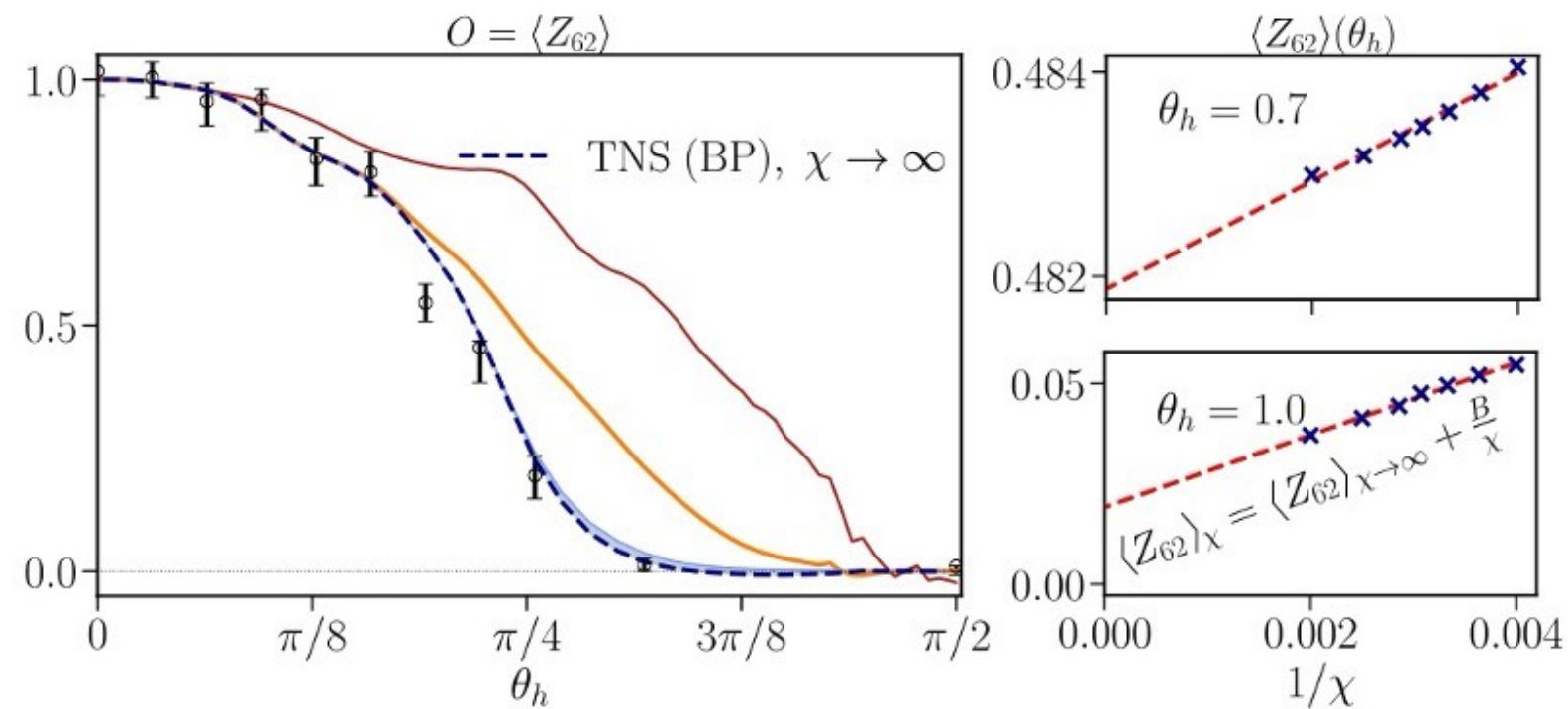
× BP results

$$O = \langle X_{13,29,31} Y_{9,30} Z_{8,12,17,28,32} \rangle$$



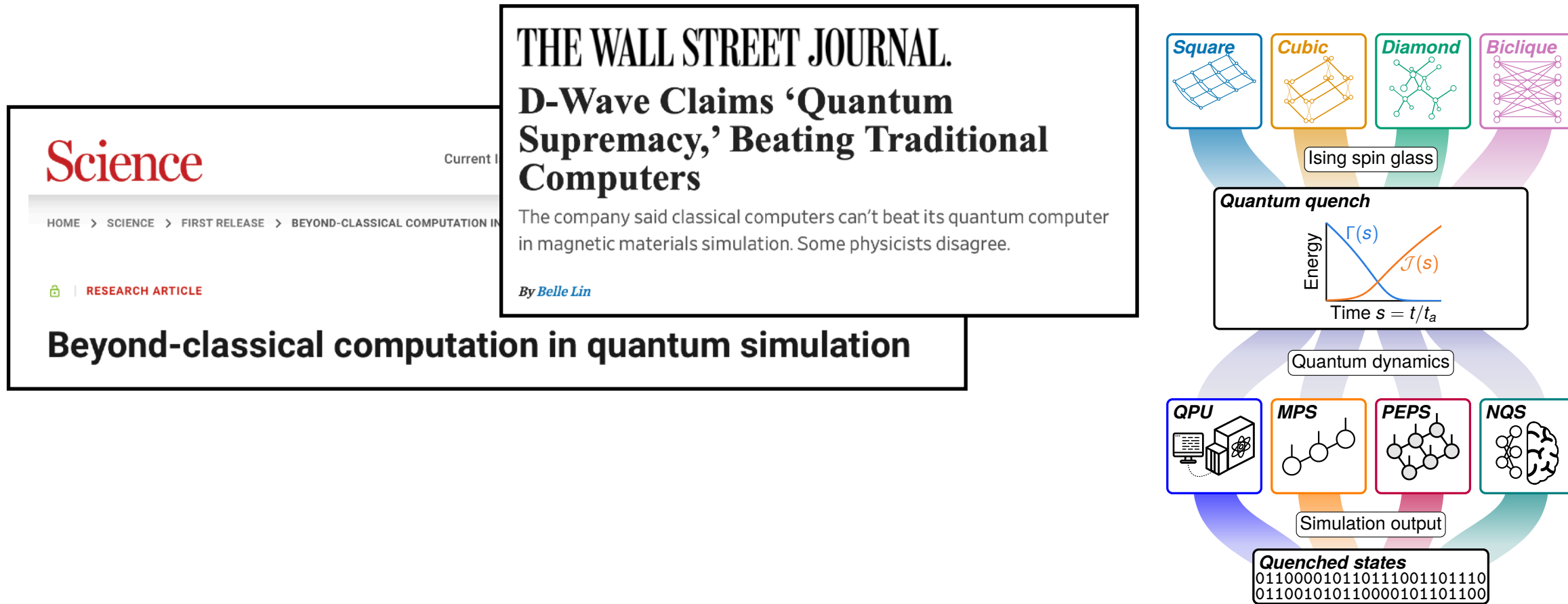
Results in
verifiable regime

--- BP results



Results in
unverifiable regime

2) Simulation of D-Wave quantum Ising dynamics



Dynamics of disordered quantum systems with two- and three-dimensional tensor networks

Joseph Tindall,¹ Antonio Francesco Mello,^{1,2} Matthew Fishman,¹ E. Miles Stoudenmire,¹ and Dries Sels^{1,3}

¹Center for Computational Quantum Physics, Flatiron Institute, New York, New York 10010, USA

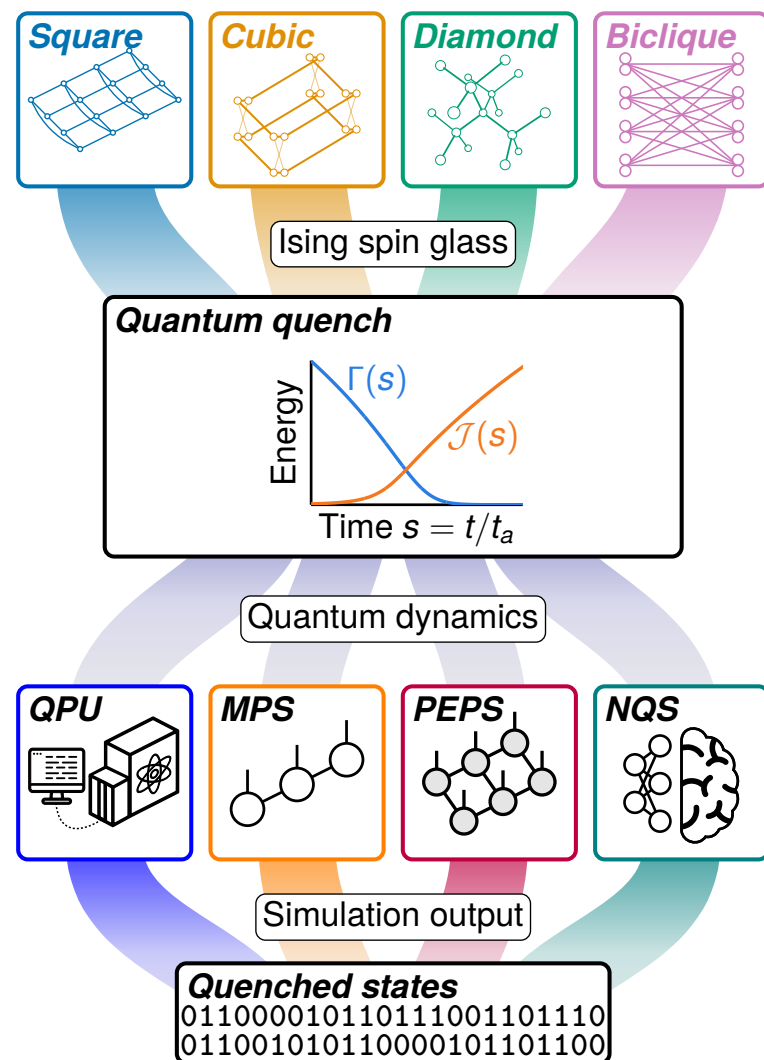
²International School for Advanced Studies (SISSA), via Bonomea 265, 34136 Trieste, Italy

³Center for Quantum Phenomena, Department of Physics,
New York University, 726 Broadway, New York, NY, 10003, USA

(Dated: March 14, 2025)

2) Simulation of D-Wave quantum Ising dynamics

What did the experiment compute?

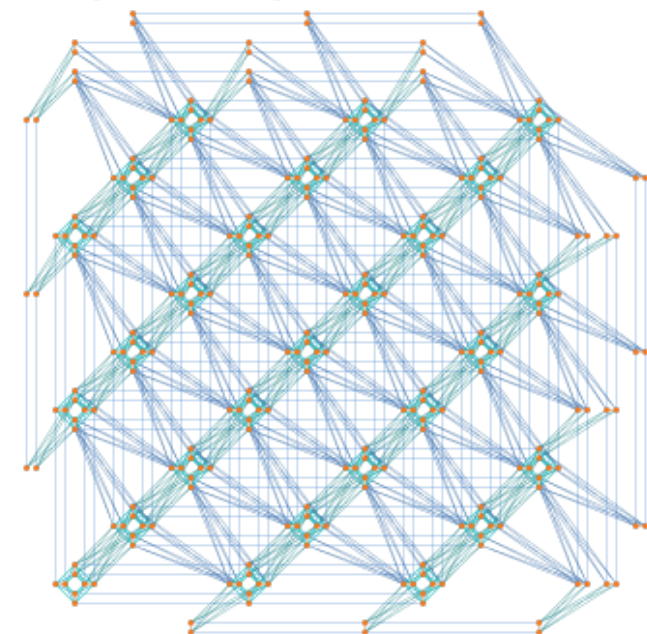


We consider a time-dependent Hamiltonian that interpolates between a driving Hamiltonian $\mathcal{H}_D$ and a classical Ising problem Hamiltonian $\mathcal{H}_P$:

$$\mathcal{H}(t) = \Gamma(t/t_a)\mathcal{H}_D + \mathcal{J}(t/t_a)\mathcal{H}_P, \quad (1)$$

$$\mathcal{H}_D = - \sum_i \sigma_i^x, \quad \mathcal{H}_P = \sum_{i<j} J_{ij} \sigma_i^z \sigma_j^z, \quad (2)$$

Hardware: SQUIDs (quantum Ising spins) on "Pegasus" graph topology



Goal: compute **post-quench correlations**

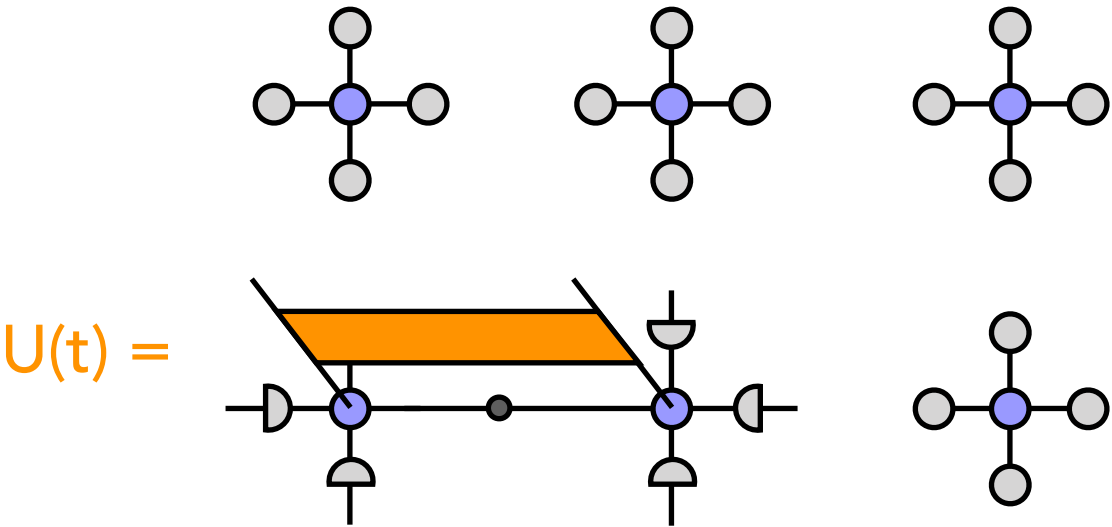
$$c_{ij} = \langle \sigma_i^z \sigma_j^z \rangle \quad \epsilon_c = \left(\frac{\sum_{i,j} (c_{ij} - \tilde{c}_{ij})^2}{\sum_{i,j} \tilde{c}_{ij}^2} \right)^{1/2}$$

Tensor BP for Annealing

Our approach to simulate:

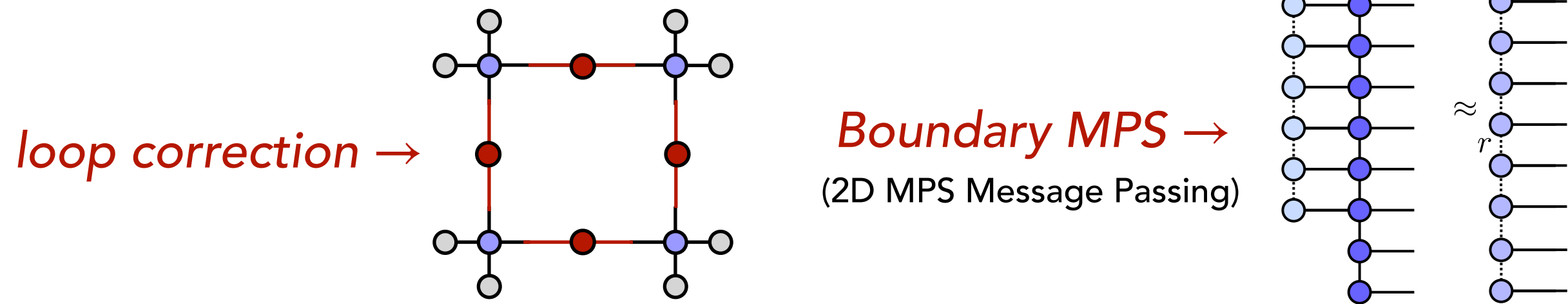
1. Time evolution at BP level

(allowing bonds to grow large, cheaply)



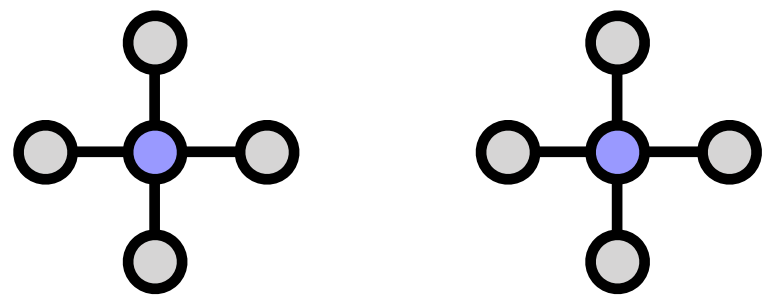
2. Corrected calculation of correlators

(truncate bonds before this step)

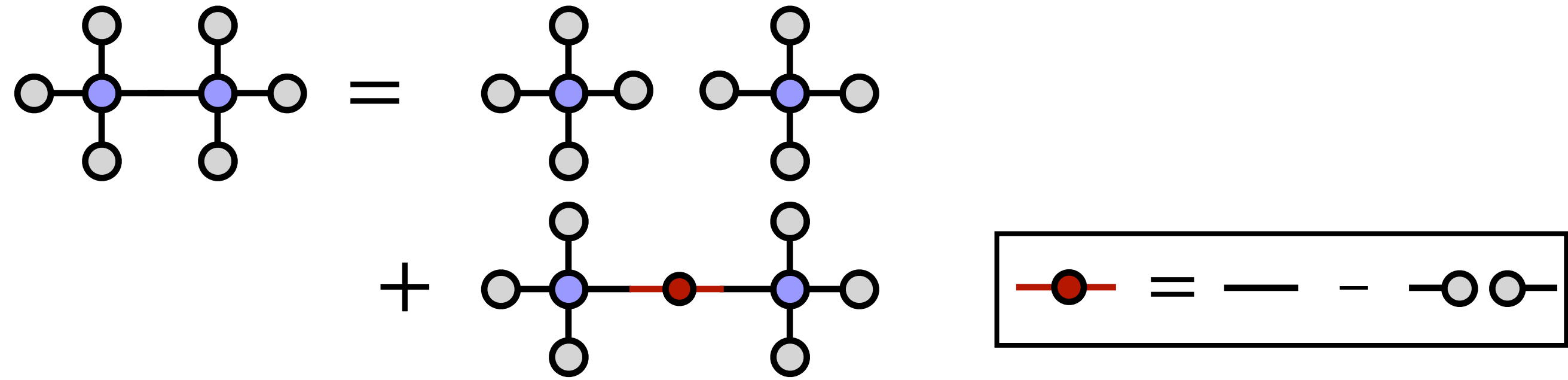


Loop Corrections – What are They?

Messages "cut" the original network:

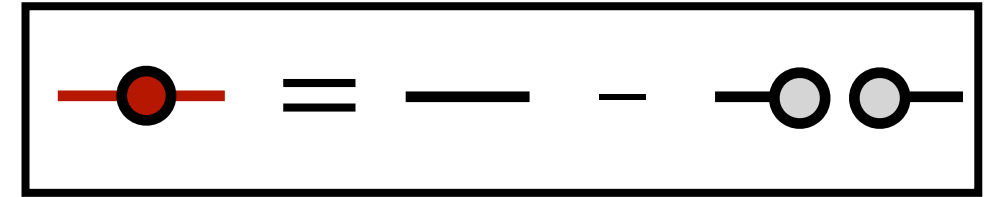


Missing contributions from orthogonal "message space":

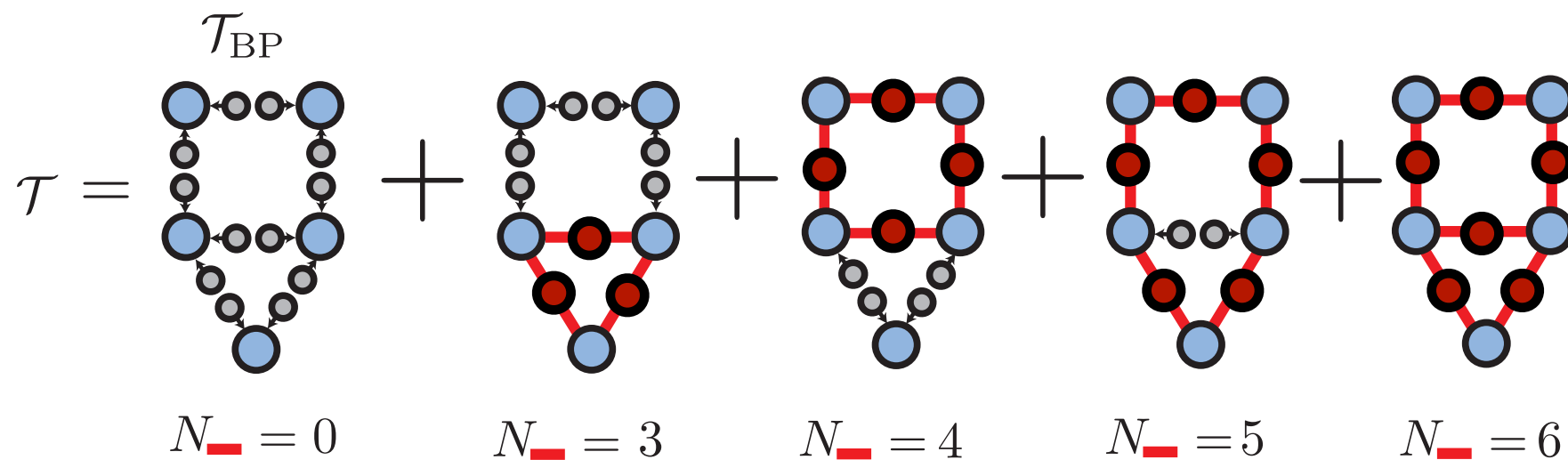


Only **closed loops** give non-zero correction

Loop Corrections – What are They?



Only **closed loops** give non-zero correction



Like a Feynman diagram expansion but controlled by correlation length

Glen Evenbly, et al., "Loop Series Expansions for Tensor Networks", Phys. Rev. Research 8 1 (2026)

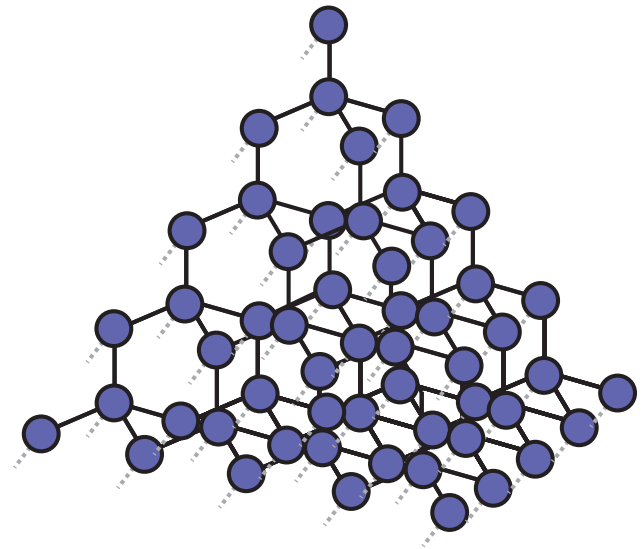
Siddhant Midha and Yifan F. Zhang, "Beyond Belief Propagation...." PRX Quantum 7, 033010 (2026)

J. Gray et al, "Tensor Network Loop Cluster Expansions....." PRB 113 235135 (2026)

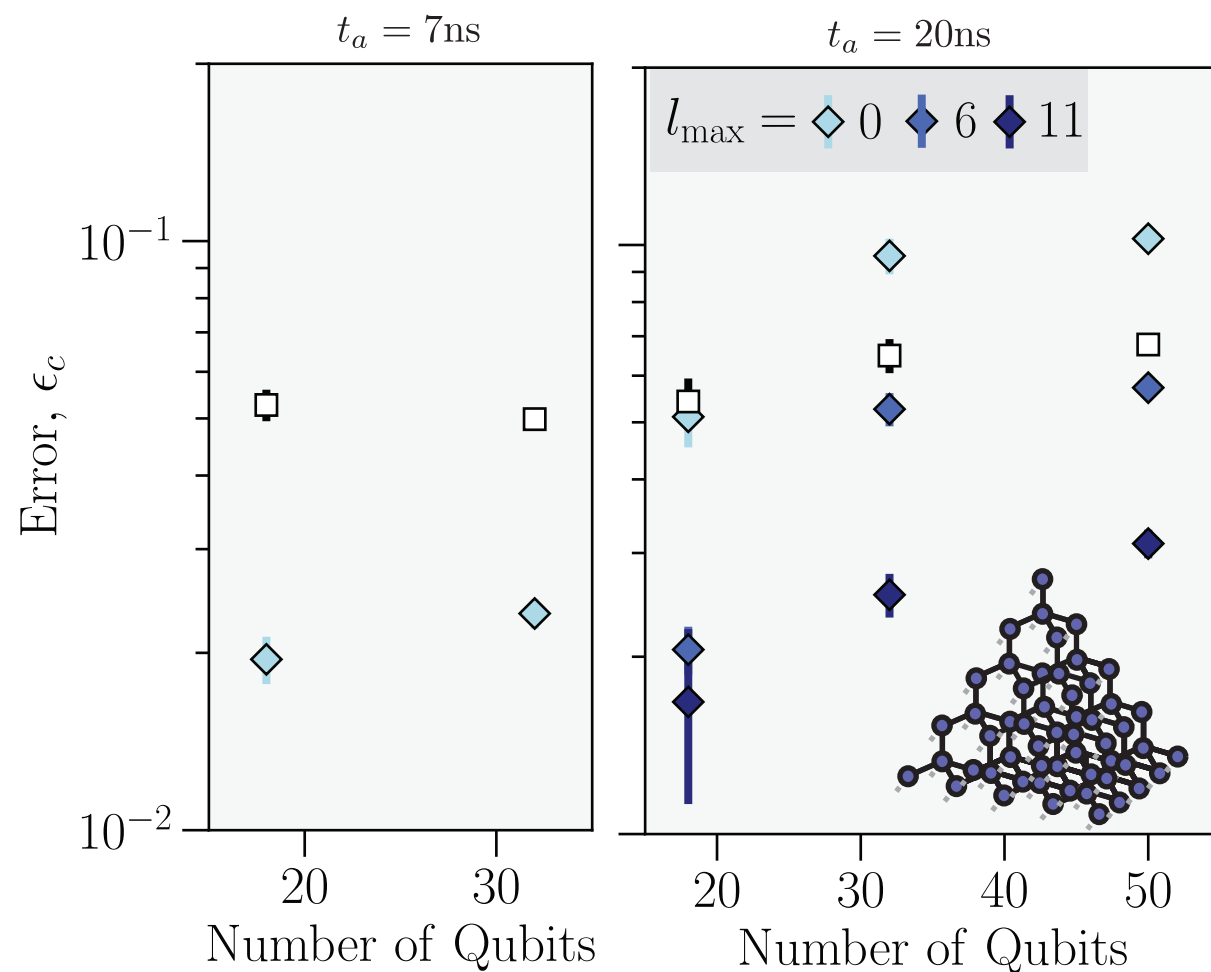
Annealing Simulation Results

Can compute **3D** quantum dynamics with these tools

Diamond Cubic Tensor Network $|\psi\rangle$



Correlator errors:



Quantum annealer

Loop corrected BP method

Competitive with quantum annealer results,
using scalable methods

Annealing Simulation Results

See the nice Physics Today blog on this:



[Home](#) » [Quantum](#) » [Quantum computing](#) » No quantum advantage (yet) in the world of tensor networks

QUANTUM COMPUTING | RESEARCH UPDATE

No quantum advantage (yet) in the world of tensor networks

21 Aug 2026 [Nohora Hernández](#)

Reading Time: 4 minutes



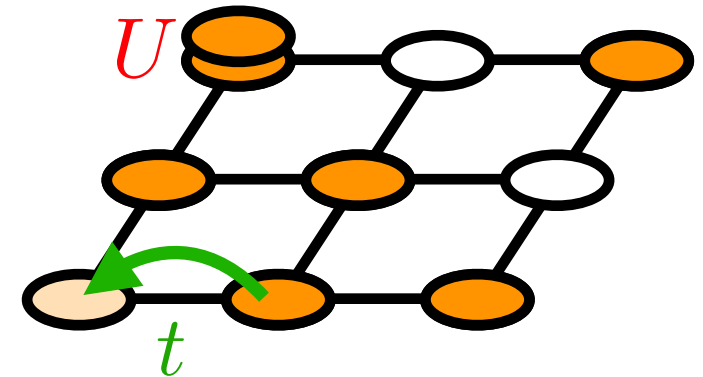
Tensor networks pictographic scheme

Tensor networks pictographic scheme Each tensor is represented by an array of dimension N , that matches its number of indices. By joining the legs (indices) of each tensor, it is possible to encode the Hamiltonian or the state of a many-particle system. (Courtesy: Lucy Reading-Ikkanda/Simons Foundation)

Future Directions & Open Questions

What is the limit?

Can we simulate dynamics of strongly-correlated electrons in 2D and 3D?



$$\hat{H} = -t \sum_{ij} (\hat{c}_i^\dagger \hat{c}_j + \hat{c}_j^\dagger \hat{c}_i) + U \sum_j \hat{n}_{j\uparrow} \hat{n}_{j\downarrow}$$

Can novel dynamics approaches (space + time)

"influence functional [1]" bring long time dynamics into reach?

Summary

Newly **efficient** methods for 2D and 3D tensor networks are being developed based on message passing

Actually easier in **high dimensions**
(more neighbors)

Opportunities to study **quantum dynamics** in higher dimensions

Now we are going to work with our own BP + PEPS Code using **ITensors.jl** and **NamedGraphs.jl**